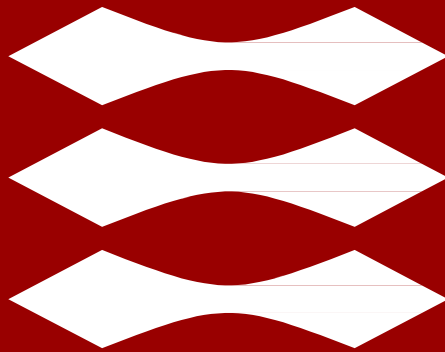


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Work, Energy, and Power



10060 Physics (Polytechnical Foundation)

Work

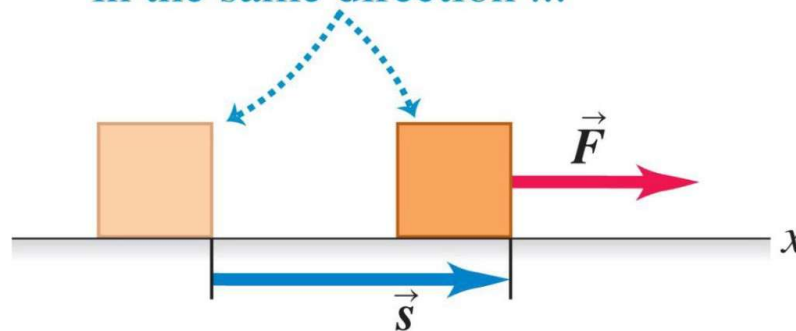
How much energy do you use when you deadlift?



(Discuss in groups)

Work: Definition

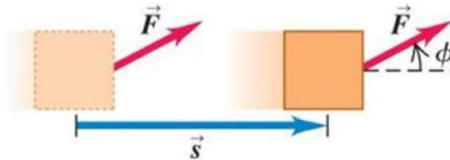
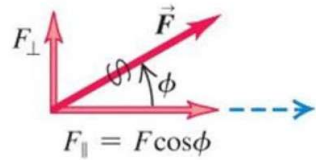
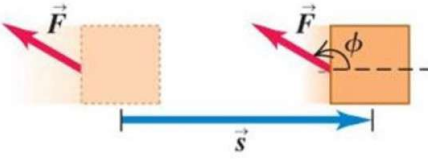
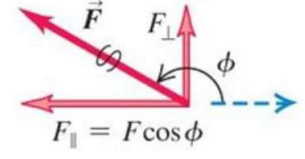
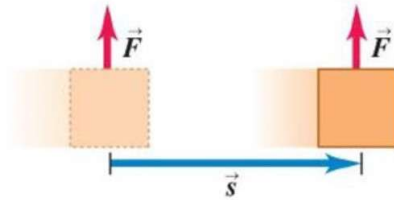
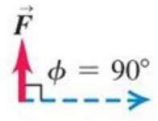
If a particle moves through a displacement $\vec{s}$ while a constant force $\vec{F}$ acts on it in the same direction ...



$$W = \vec{F} \cdot \Delta \vec{r}$$

... the work done by the force on the particle is $W = Fs$.

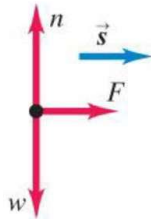
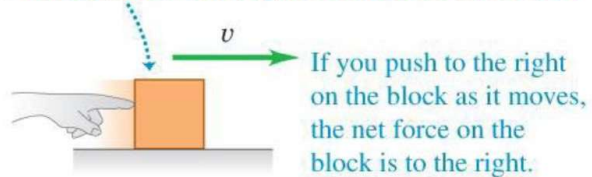
Work: Examples

Direction of Force (or Force Component)	Situation	Force Diagram
(a) Force $\vec{F}$ has a component in direction of displacement: $W = F_{\parallel}s = (F \cos \phi)s$ Work is <i>positive</i>.		
(b) Force $\vec{F}$ has a component opposite to direction of displacement: $W = F_{\parallel}s = (F \cos \phi)s$ Work is <i>negative</i> (because $F \cos \phi$ is negative for $90^\circ < \phi < 180^\circ$).		
(c) Force $\vec{F}$ (or force component $F_{\perp}$) is perpendicular to direction of displacement: The force (or force component) does <i>no</i> work on the object.		

Work: Examples

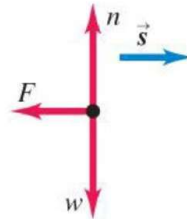
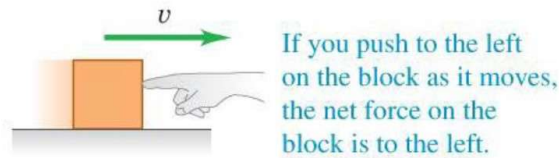
(a)

A block slides to the right on a frictionless surface.



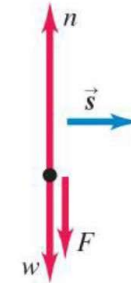
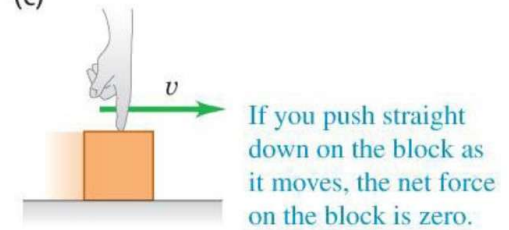
- The total work done on the block during a displacement $\vec{s}$ is positive: $W_{\text{tot}} > 0$.
- The block speeds up.

(b)



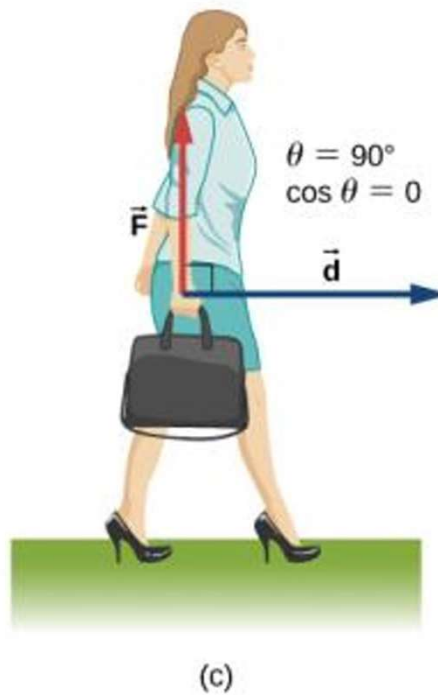
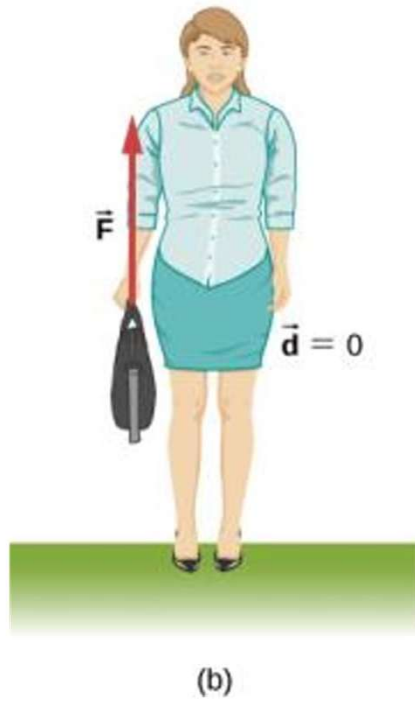
- The total work done on the block during a displacement $\vec{s}$ is negative: $W_{\text{tot}} < 0$.
- The block slows down.

(c)



- The total work done on the block during a displacement $\vec{s}$ is zero: $W_{\text{tot}} = 0$.
- The block's speed stays the same.

Work: Examples



How much work is the person doing?

How much energy do you use when you deadlift?



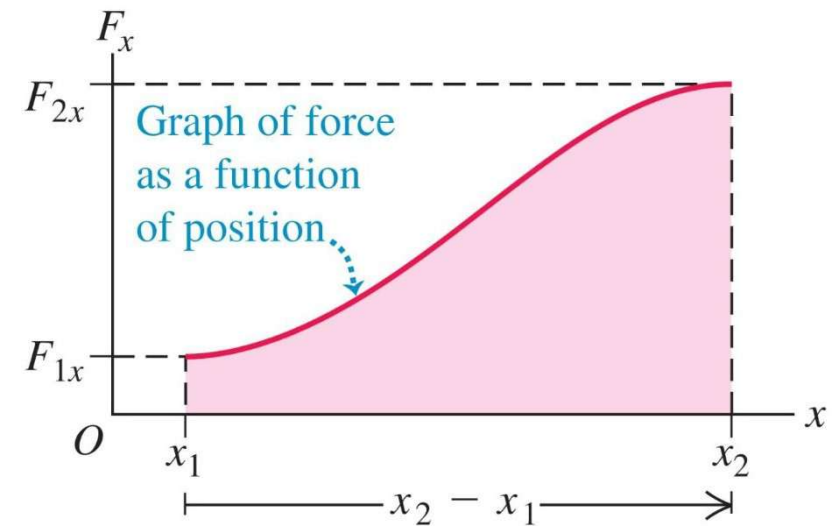
Let's calculate it

Work for a non uniform force

(a) A particle moves from x_1 to x_2 in response to a changing force in the x -direction.



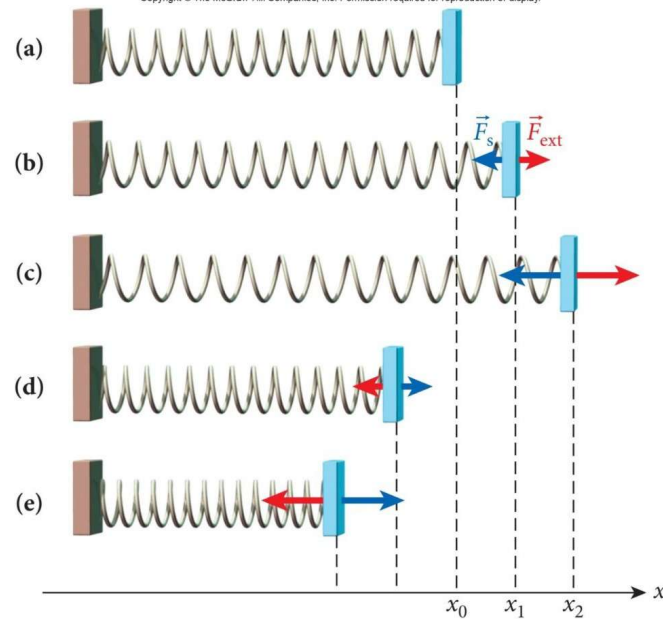
(b) The force F_x varies with position x ...



Work for non-uniform force: spring force

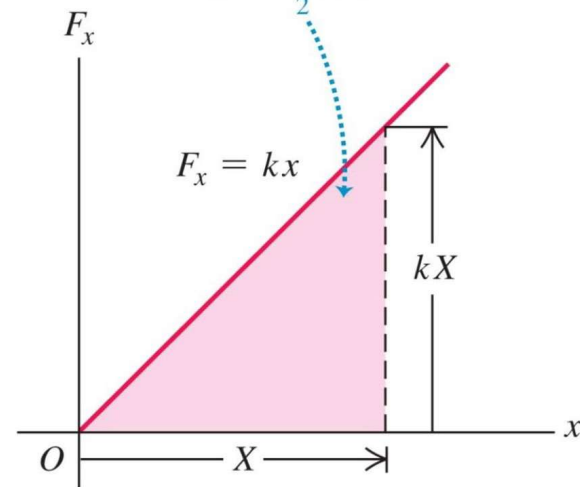
Hooke's law

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The area under the graph represents the work done on the spring as the spring is stretched from $x = 0$ to a maximum value X :

$$W = \frac{1}{2}kX^2$$



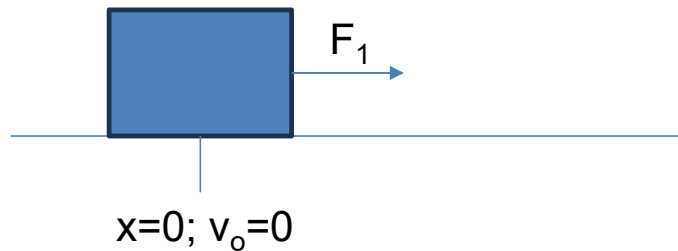
You are pushing a large box across a friction-free horizontal floor by applying a constant horizontal force. Starting from rest, you must do an amount of work W_1 to make the box travel a distance d in a time t .

How much work, W_2 , must you do (express your answer in terms of W_1) to make the box travel the same distance d in a time $3t$?

- A) $W_2 = W_1/81$
- B) $W_2 = W_1/9$
- C) $W_2 = W_1/3$
- D) $W_2 = W_1$
- E) $W_2 = 3 W_1$
- F) $W_2 = 9 W_1$
- G) $W_2 = 81 W_1$

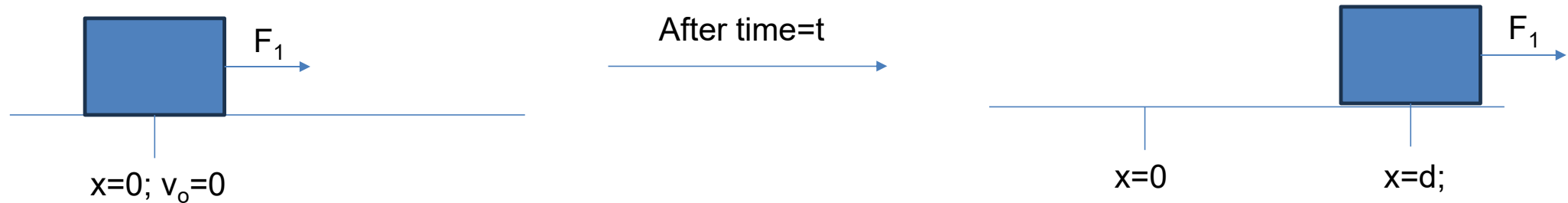
You are pushing a large box across a friction-free horizontal floor by applying a constant horizontal force, F_1 . Starting from rest, you must do an amount of work W_1 to make the box travel a distance d in a time t .

Starting also from rest, how much work, W_2 , must you do (express your answer in terms of W_1) by applying a constant horizontal force, F_2 to make the box travel the same distance d in a time $3t$?



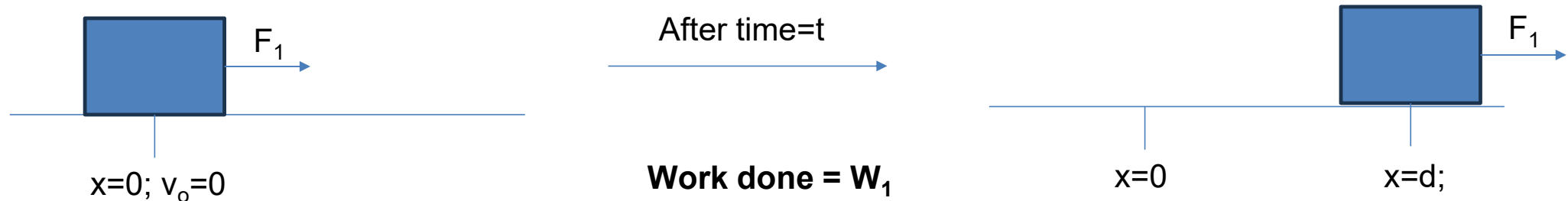
You are pushing a large box across a friction-free horizontal floor by applying a constant horizontal force, F_1 . Starting from rest, you must do an amount of work W_1 to make the box travel a distance d in a time t .

Starting also from rest, how much work, W_2 , must you do (express your answer in terms of W_1) by applying a constant horizontal force, F_2 to make the box travel the same distance d in a time $3t$?



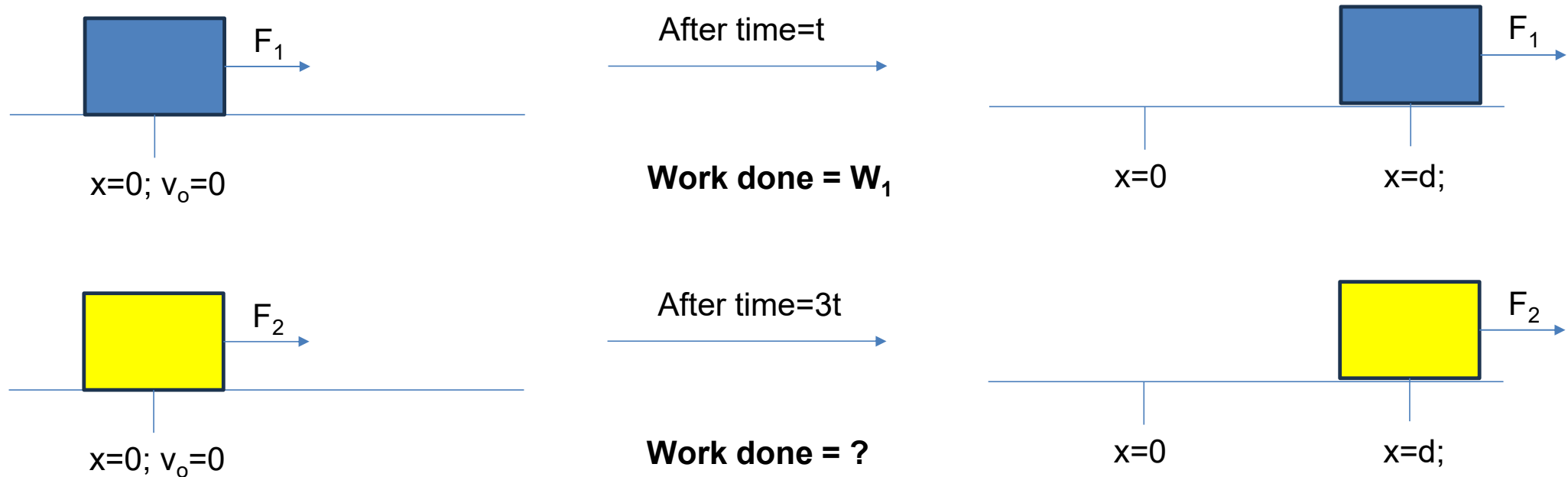
You are pushing a large box across a friction-free horizontal floor by applying a constant horizontal force, F_1 . Starting from rest, you must do an amount of work W_1 to make the box travel a distance d in a time t .

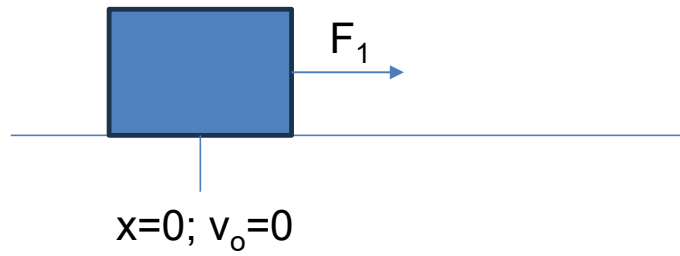
Starting also from rest, how much work, W_2 , must you do (express your answer in terms of W_1) by applying a constant horizontal force, F_2 to make the box travel the same distance d in a time $3t$?



You are pushing a large box across a friction-free horizontal floor by applying a constant horizontal force, F_1 . Starting from rest, you must do an amount of work W_1 to make the box travel a distance d in a time t .

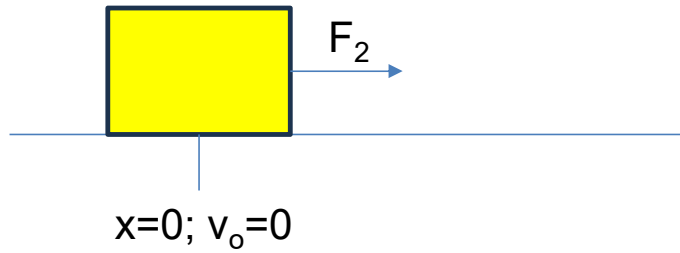
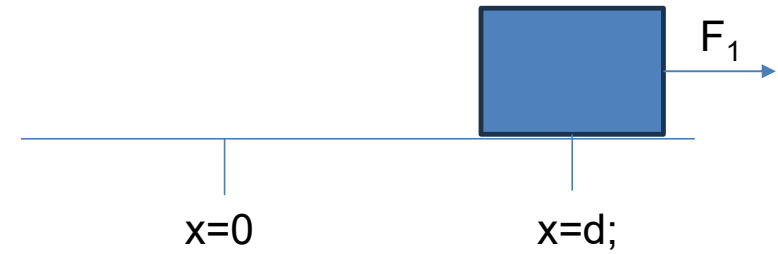
Starting also from rest, how much work, W_2 , must you do (express your answer in terms of W_1) by applying a constant horizontal force, F_2 to make the box travel the same distance d in a time $3t$?





After time= t

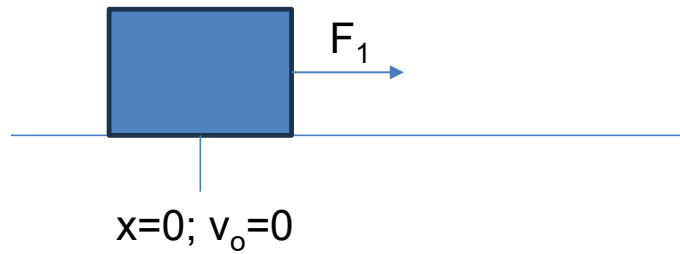
Work done = W_1



After time= $3t$

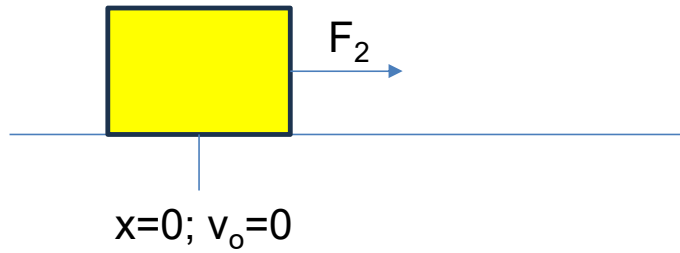
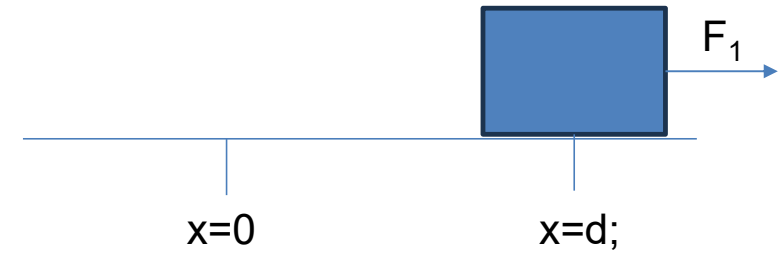
Work done = ?





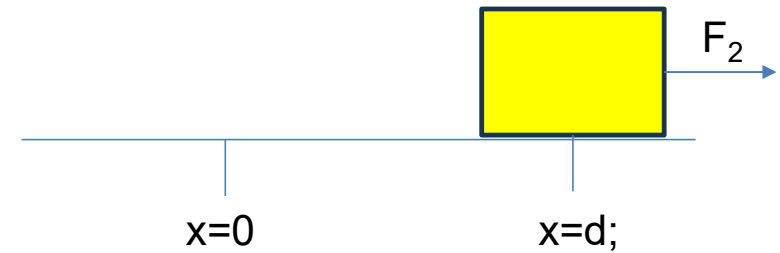
After time= t

Work done = W_1

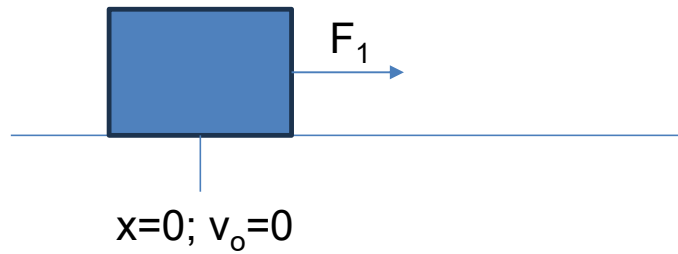


After time= $3t$

Work done = $W_2?$

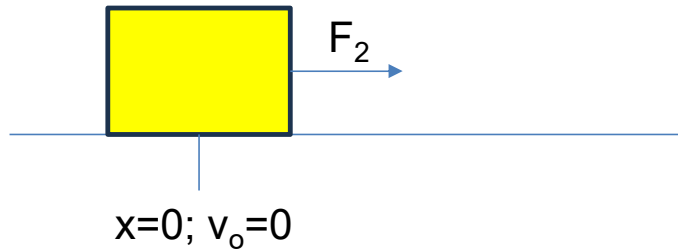
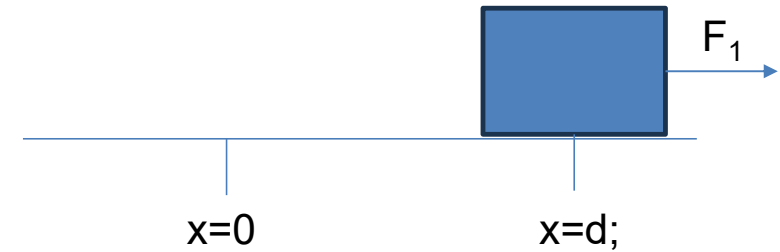


Work is $W = F \cdot d$



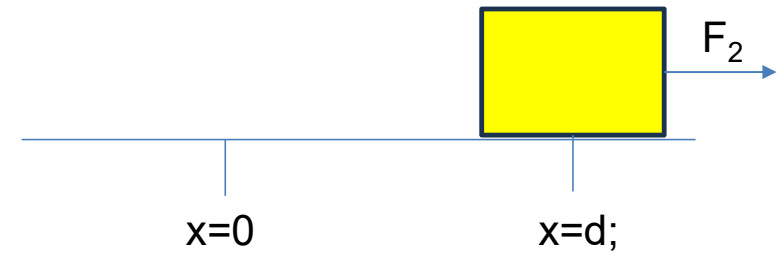
After time= t

Work done = W_1

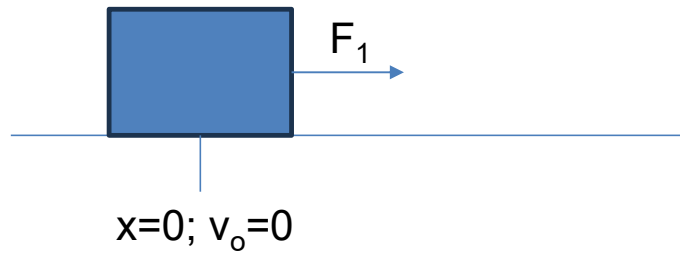


After time= $3t$

Work done = $W_2?$

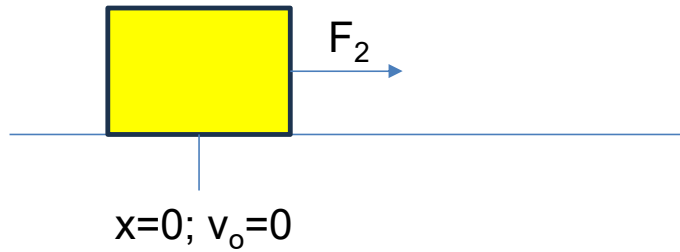
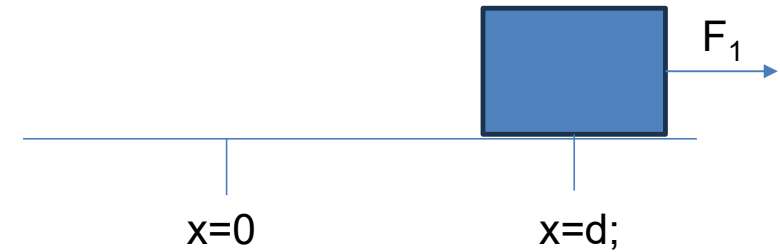


$$\text{Work is } W = F \cdot d \rightarrow \frac{W_1}{W_2} = \frac{F_1 \cdot d}{F_2 \cdot d}$$



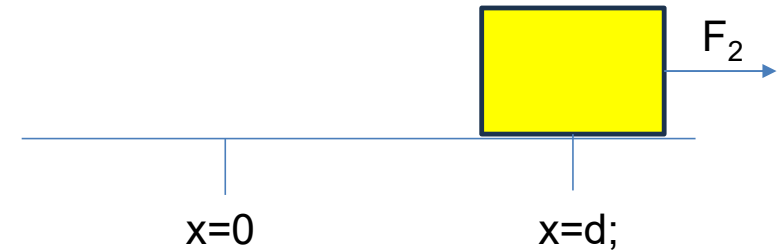
After time= t

Work done = W_1



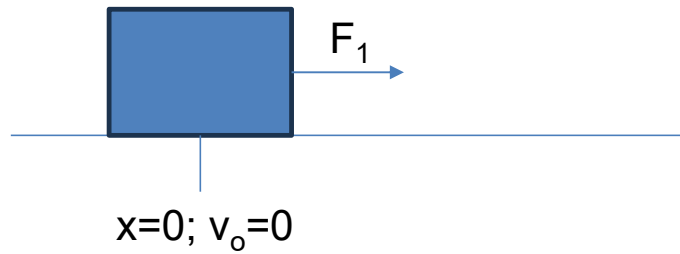
After time= $3t$

Work done = ?



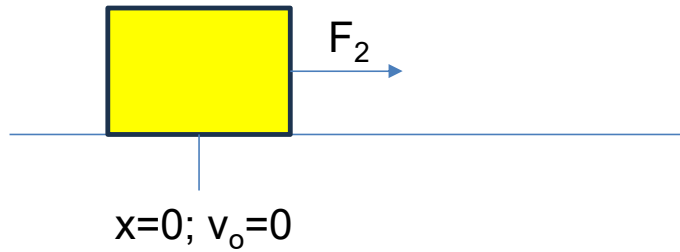
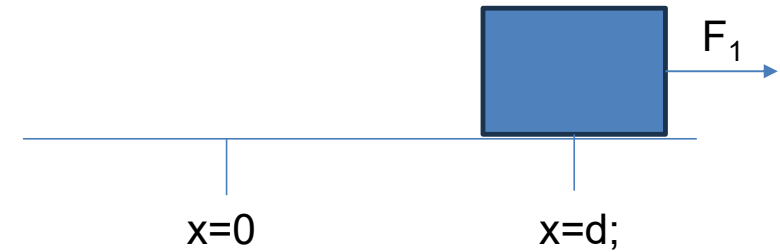
$$\text{Work is } W = F \cdot d \rightarrow \frac{W_1}{W_2} = \frac{F_1 \cdot d}{F_2 \cdot d}$$

d is the same in both cases



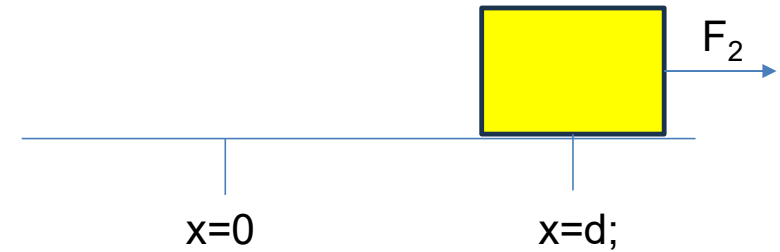
After time= t

Work done = W_1



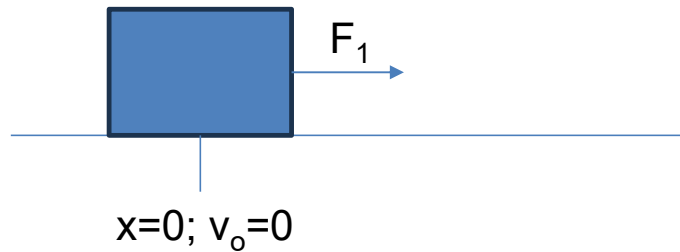
After time= $3t$

Work done = ?



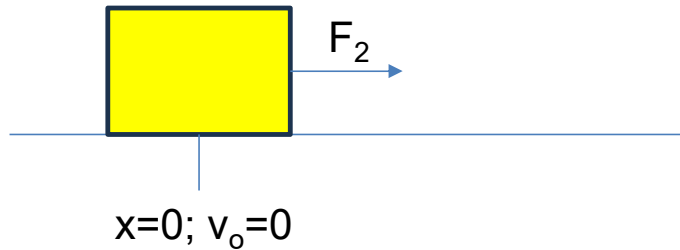
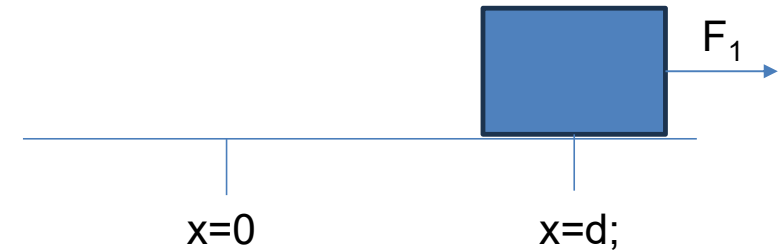
$$\text{Work is } W = F \cdot d \rightarrow \frac{W_1}{W_2} = \frac{F_1 \cdot d}{F_2 \cdot d} = \frac{F_1}{F_2}$$

d is the same in both cases



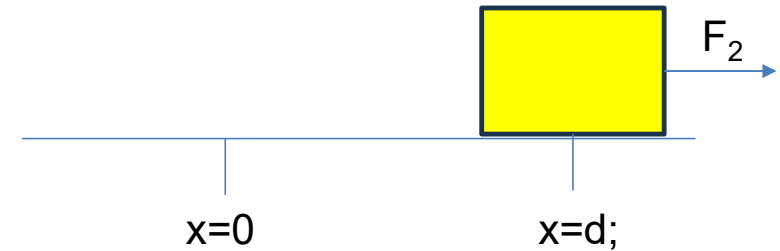
After time= t

Work done = W_1



After time= $3t$

Work done = ?

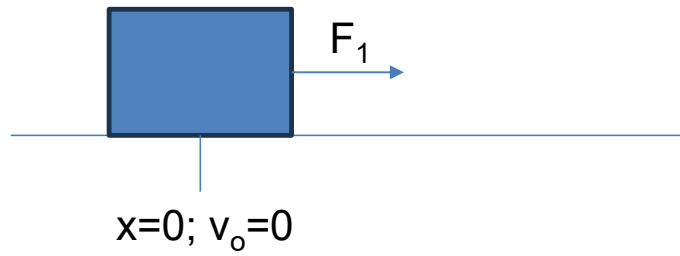


Work is $W = F \cdot d \rightarrow \frac{W_1}{W_2} = \frac{F_1 \cdot d}{F_2 \cdot d} = \frac{F_1}{F_2}$

$$\frac{W_1}{W_2} = \frac{F_1}{F_2}$$

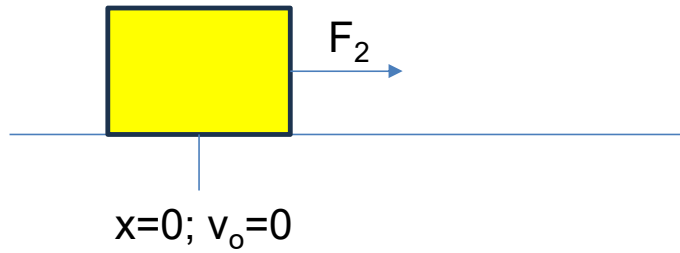
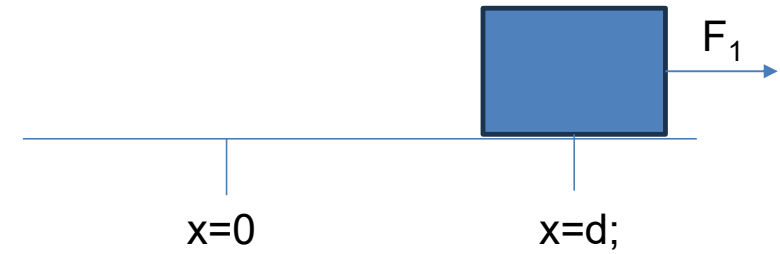
d is the same in both cases

All we need to do is figure out the relation between F_1 and F_2



After time= t

Work done = W_1

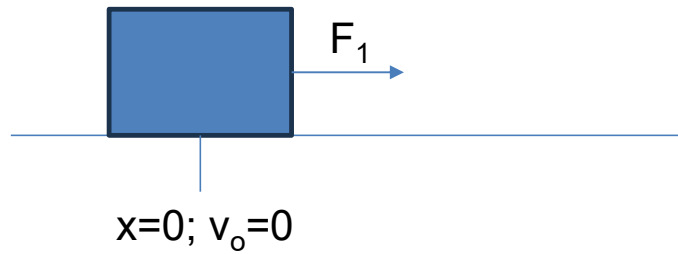


After time= $3t$

Work done = ?

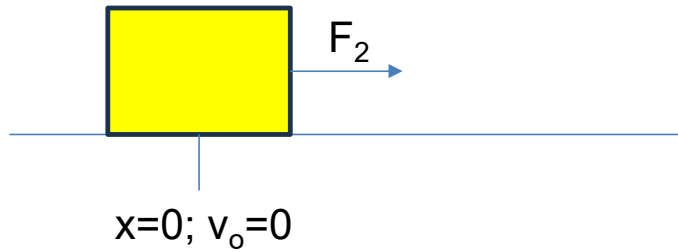
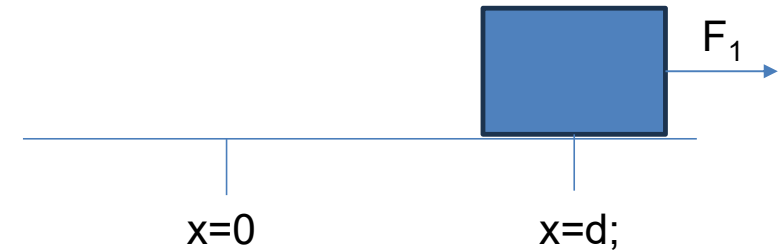


$$\frac{W_1}{W_2} = \frac{F_1}{F_2}$$



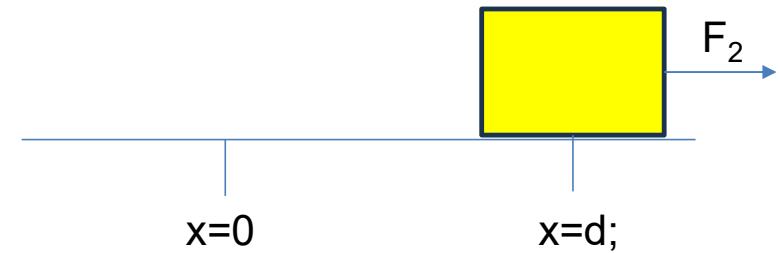
After time= t

Work done = W_1



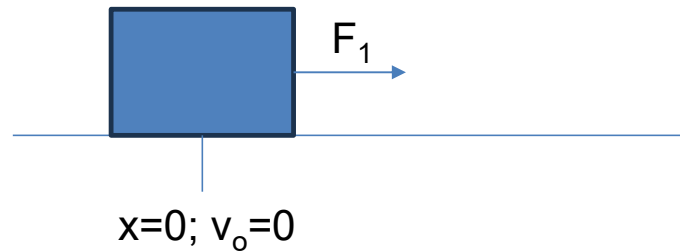
After time= $3t$

Work done = ?



$$\frac{W_1}{W_2} = \frac{F_1}{F_2}$$

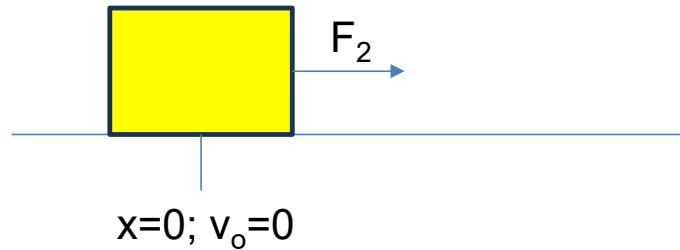
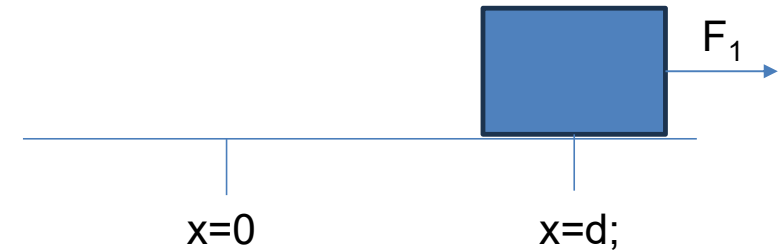
The only thing that changes between the two situations is the time.



After time= t



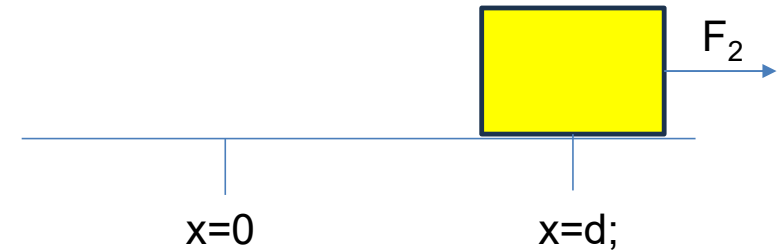
Work done = W_1



After time= $3t$

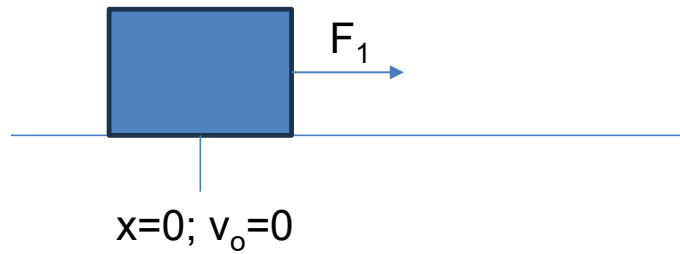


Work done = ?



$$\frac{W_1}{W_2} = \frac{F_1}{F_2}$$

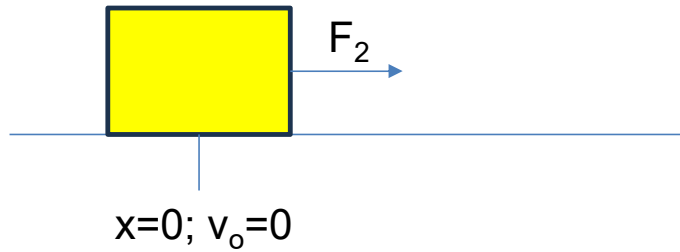
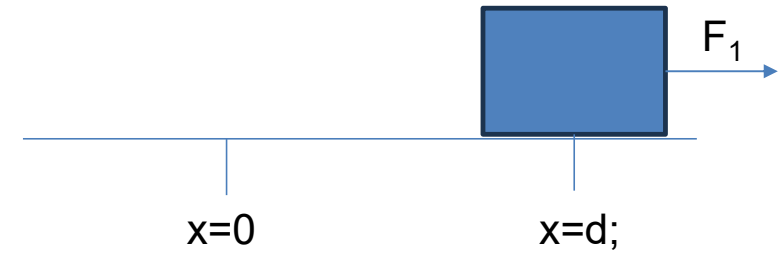
The only thing that changes between the two situations is the time.
So we need to figure out the relationship between the force applied
and the time spent between $x=0$ and $x=d$



After time= t

→

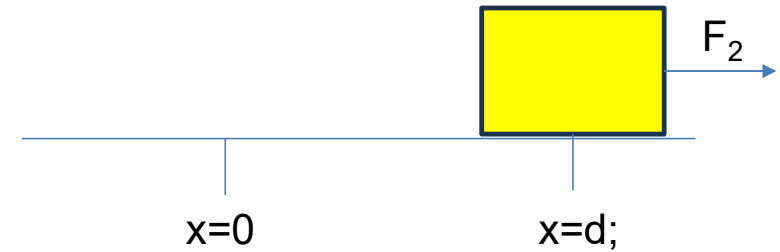
Work done = W_1



After time= $3t$

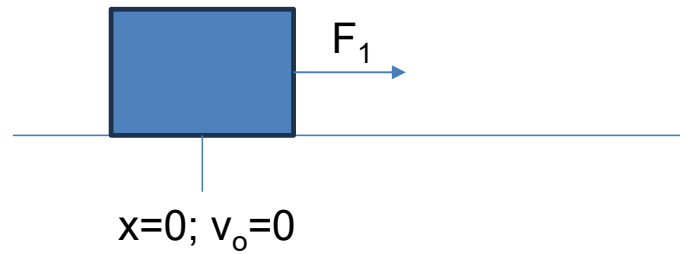
→

Work done = ?



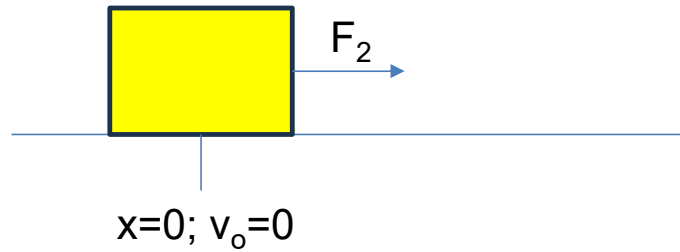
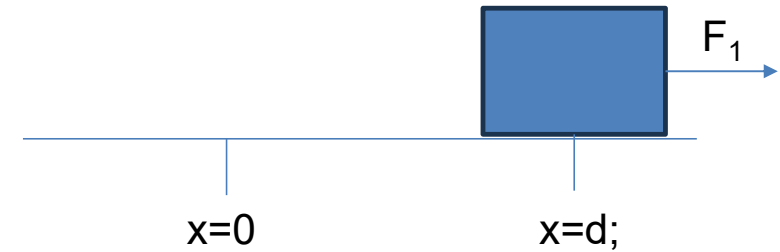
$$x = x_o + v_o \cdot t + \frac{1}{2} \cdot a \cdot t^2$$

$$\frac{W_1}{W_2} = \frac{F_1}{F_2}$$



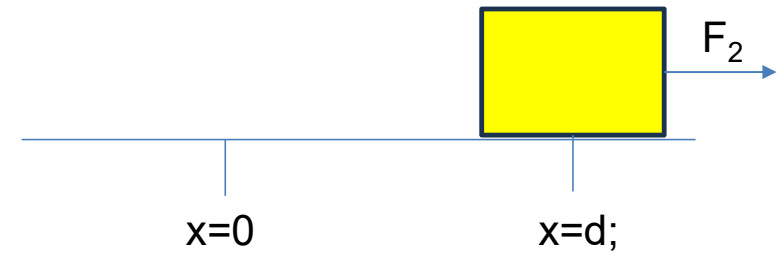
After time= t

Work done = W_1



After time= $3t$

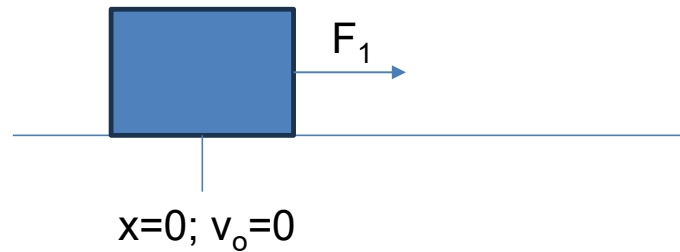
Work done = ?



$$x = x_o + v_o \cdot t + \frac{1}{2} \cdot a \cdot t^2$$

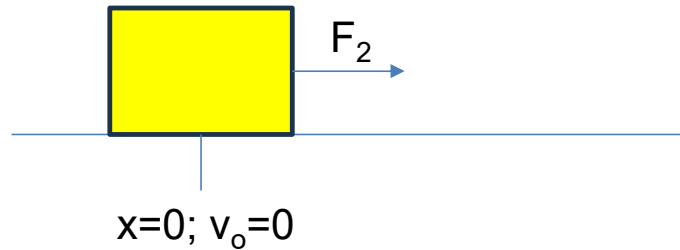
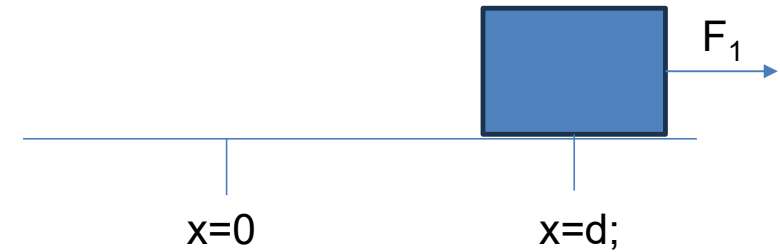
$$\frac{W_1}{W_2} = \frac{F_1}{F_2}$$

Final position is $x=d \longrightarrow d = x_o + v_o \cdot t + \frac{1}{2} \cdot a \cdot t^2$



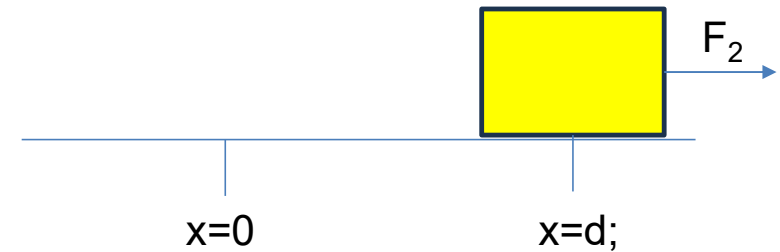
After time= t

Work done = W_1



After time= $3t$

Work done = ?

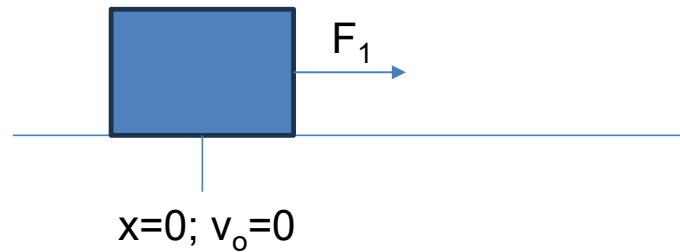


$$x = x_o + v_o \cdot t + \frac{1}{2} \cdot a \cdot t^2$$

$$\frac{W_1}{W_2} = \frac{F_1}{F_2}$$

Final position is $x=d \longrightarrow d = x_o + v_o \cdot t + \frac{1}{2} \cdot a \cdot t^2$

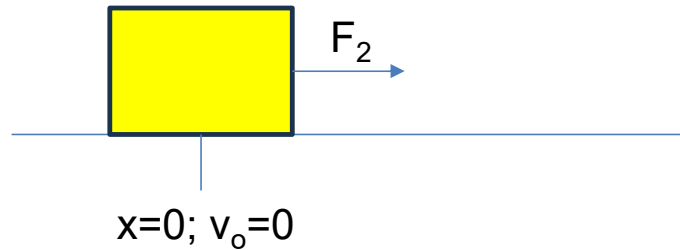
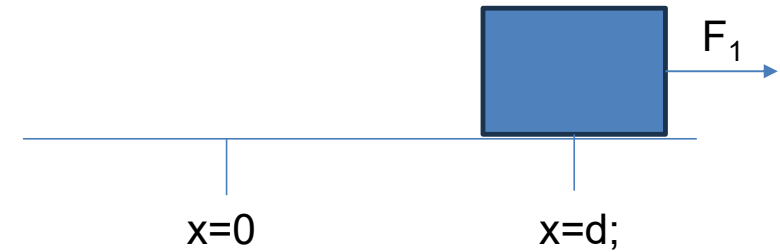
Initial position is $x_o=0 \longrightarrow d = v_o \cdot t + \frac{1}{2} \cdot a \cdot t^2$



After time= t

→

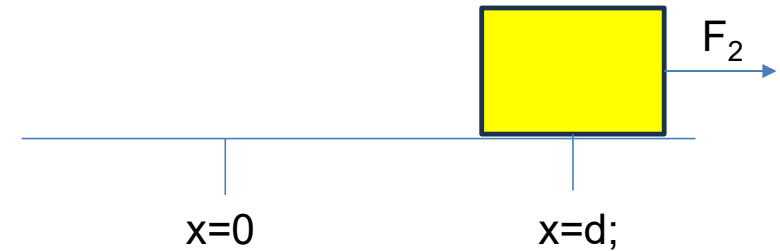
Work done = W_1



After time= $3t$

→

Work done = ?



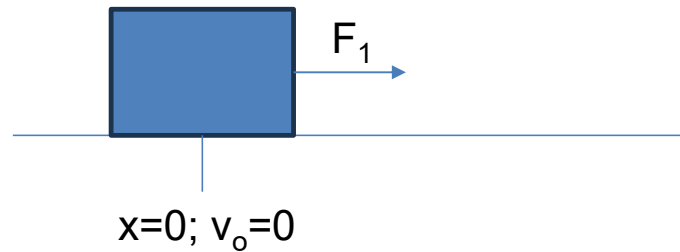
$$x = x_o + v_o \cdot t + \frac{1}{2} \cdot a \cdot t^2$$

$$\frac{W_1}{W_2} = \frac{F_1}{F_2}$$

Final position is $x=d \longrightarrow d = x_o + v_o \cdot t + \frac{1}{2} \cdot a \cdot t^2$

Initial position is $x_o=0 \longrightarrow d = v_o \cdot t + \frac{1}{2} \cdot a \cdot t^2$

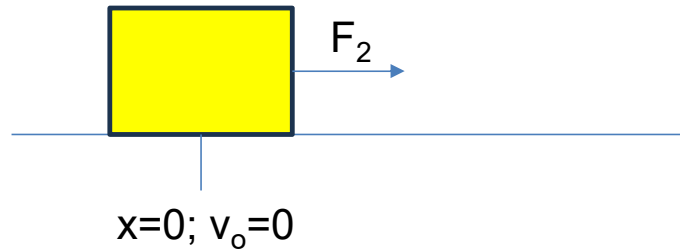
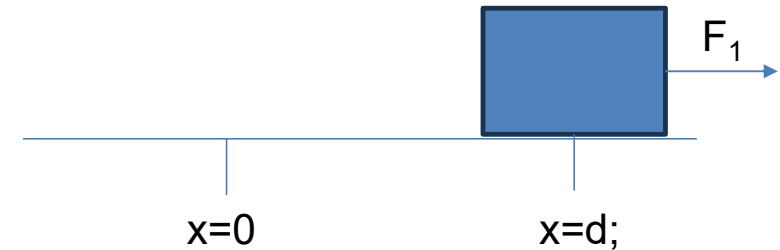
Initial speed is $v_o=0 \longrightarrow d = \frac{1}{2} \cdot a \cdot t^2$



After time= t

→

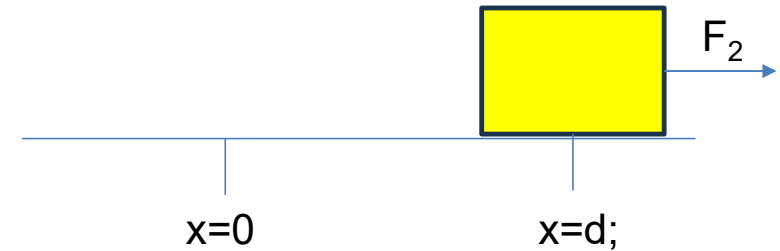
Work done = W_1



After time= $3t$

→

Work done = ?



$$x = x_o + v_o \cdot t + \frac{1}{2} \cdot a \cdot t^2$$

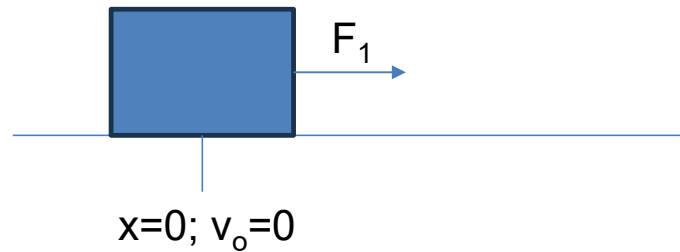
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Final position is $x=d \longrightarrow d = x_o + v_o \cdot t + \frac{1}{2} \cdot a \cdot t^2$

Initial position is $x_o=0 \longrightarrow d = v_o \cdot t + \frac{1}{2} \cdot a \cdot t^2$

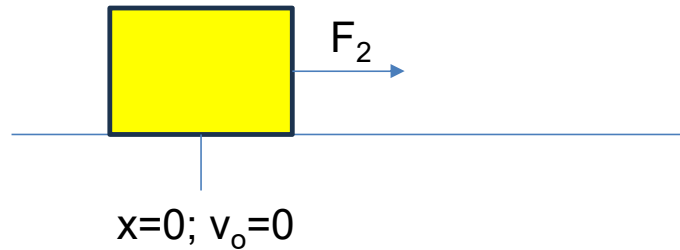
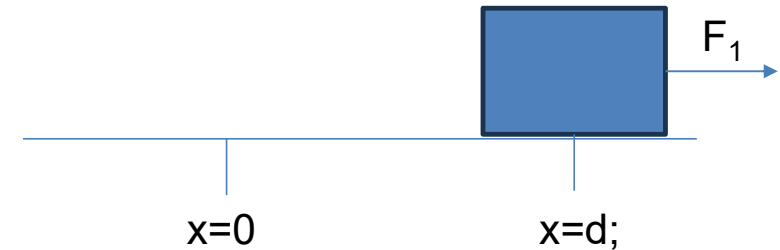
Initial speed is $v_o=0 \longrightarrow d = \frac{1}{2} \cdot a \cdot t^2$

Newton's law:
 $F = m \cdot a; a = \frac{F}{m}$



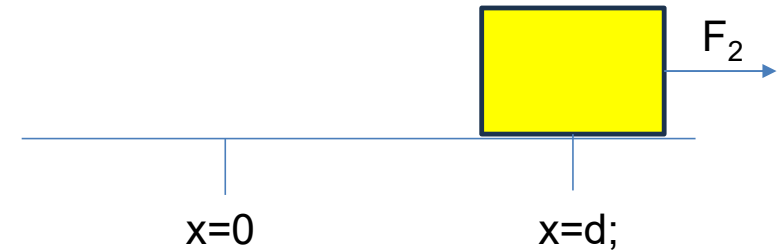
After time= t

Work done = W_1



After time= $3t$

Work done = ?



$$x = x_o + v_o \cdot t + \frac{1}{2} \cdot a \cdot t^2$$

$$\frac{W_1}{W_2} = \frac{F_1}{F_2}$$

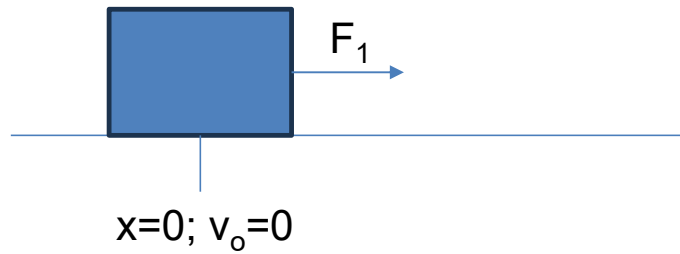
Final position is $x=d \longrightarrow d = x_o + v_o \cdot t + \frac{1}{2} \cdot a \cdot t^2$

Initial position is $x_o=0 \longrightarrow d = v_o \cdot t + \frac{1}{2} \cdot a \cdot t^2$

Initial speed is $v_o=0 \longrightarrow d = \frac{1}{2} \cdot a \cdot t^2 \longrightarrow$

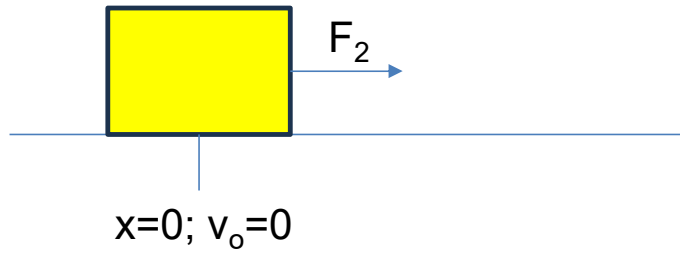
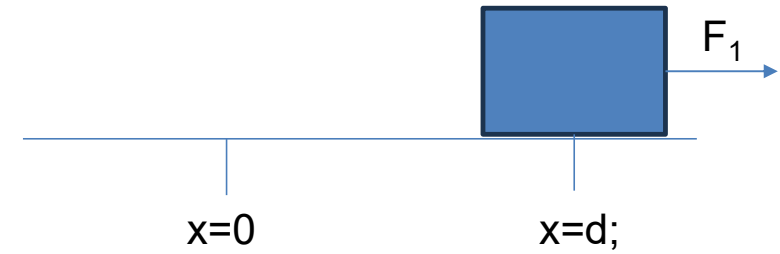
Newton's law:
 $F = m \cdot a; a = \frac{F}{m}$

$$d = \frac{1}{2} \cdot \frac{F}{m} \cdot t^2$$



After time= t

Work done = W_1



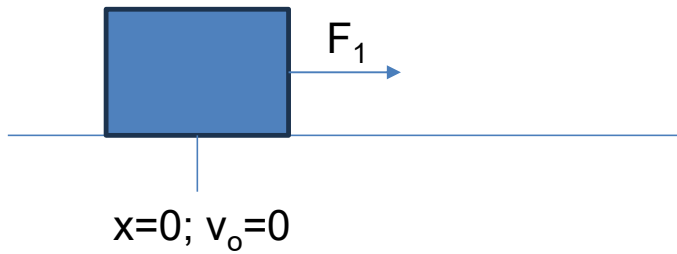
After time= $3t$

Work done = ?



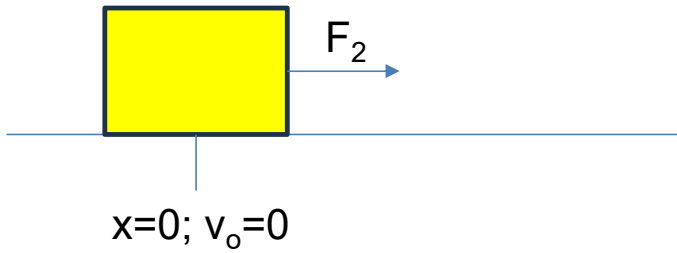
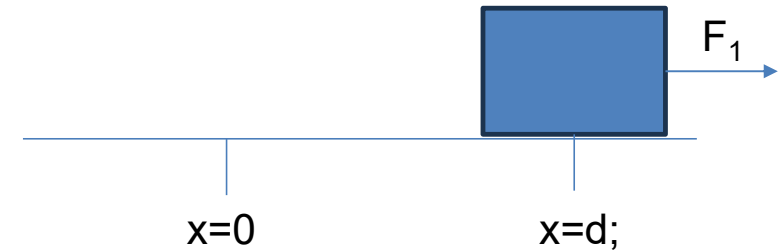
$$\frac{W_1}{W_2} = \frac{F_1}{F_2}$$

$$d = \frac{1}{2} \cdot \frac{F}{m} \cdot t^2$$



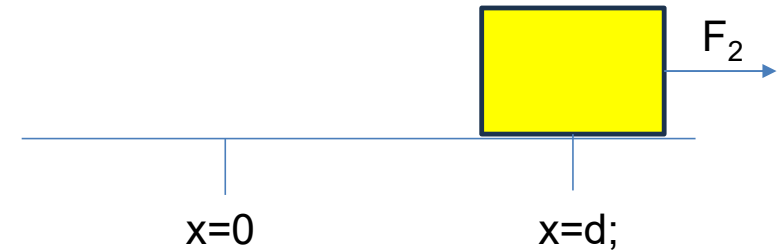
After time= t

Work done = W_1



After time= $3t$

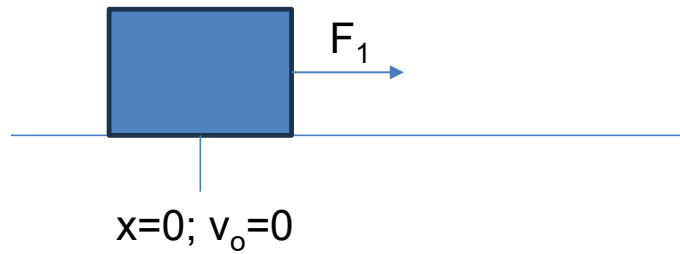
Work done = ?



$$\frac{W_1}{W_2} = \frac{F_1}{F_2}$$

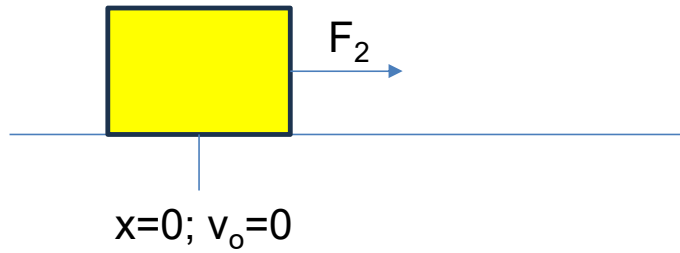
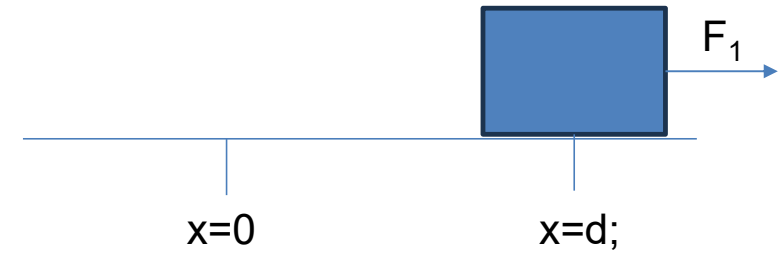
$$d = \frac{1}{2} \cdot \frac{F}{m} \cdot t^2$$

$$\frac{2 \cdot d \cdot m}{t^2} = F$$



After time= t

Work done = W_1



After time= $3t$

Work done = ?

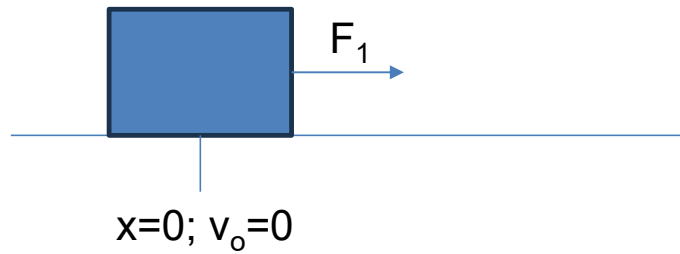


$$\frac{W_1}{W_2} = \frac{F_1}{F_2}$$

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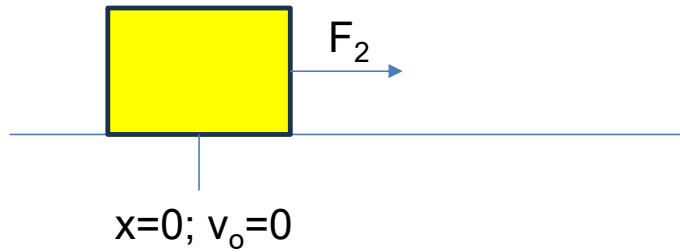
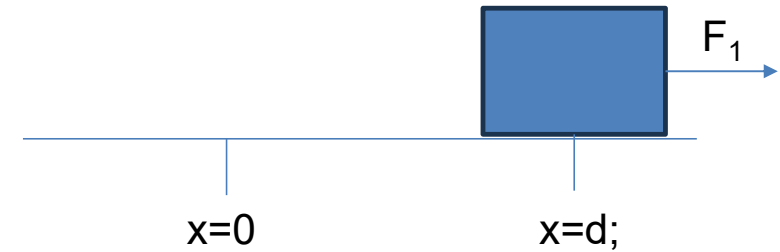
$$\frac{2 \cdot d \cdot m}{t^2} = F$$

$$\frac{W_1}{W_2} = \frac{F_1}{F_2}$$



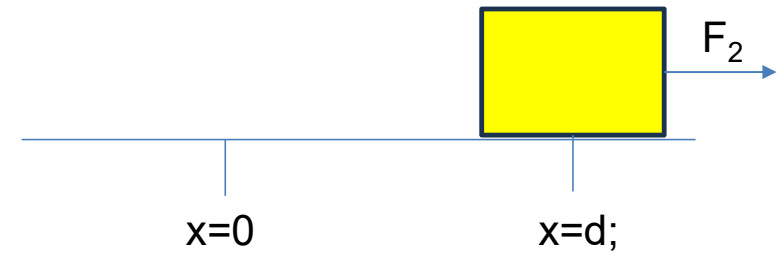
After time= t

Work done = W_1



After time= $3t$

Work done = ?

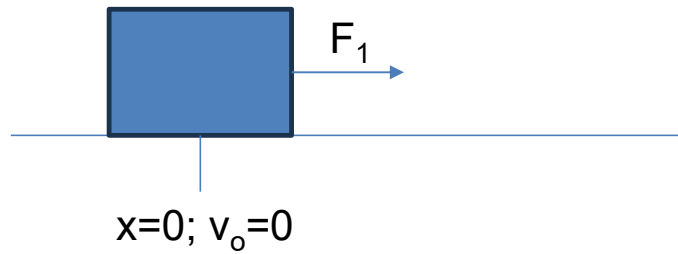


$$\frac{W_1}{W_2} = \frac{F_1}{F_2}$$

$$d = \frac{1}{2} \cdot \frac{F}{m} \cdot t^2$$

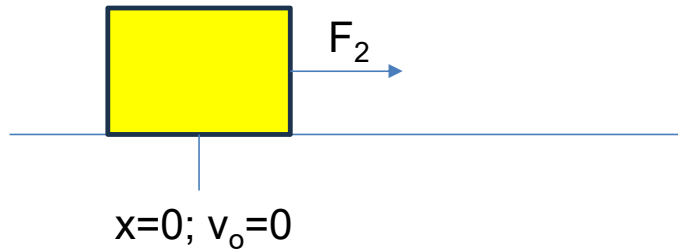
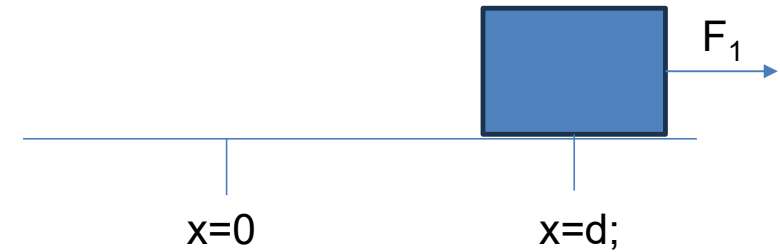
$$\frac{2 \cdot d \cdot m}{t^2} = F$$

$$\frac{W_1}{W_2} = \frac{F_1}{F_2} = \frac{\frac{2 \cdot d \cdot m}{t^2}}{\frac{2 \cdot d \cdot m}{(3t)^2}}$$



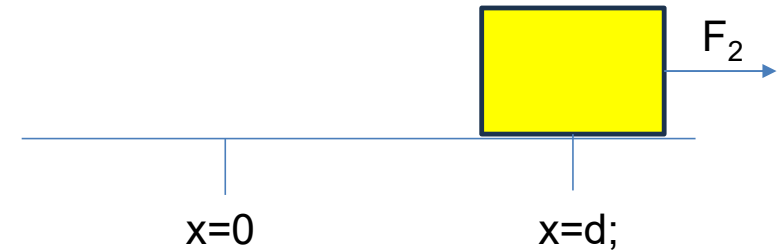
After time= t

Work done = W_1



After time= $3t$

Work done = ?

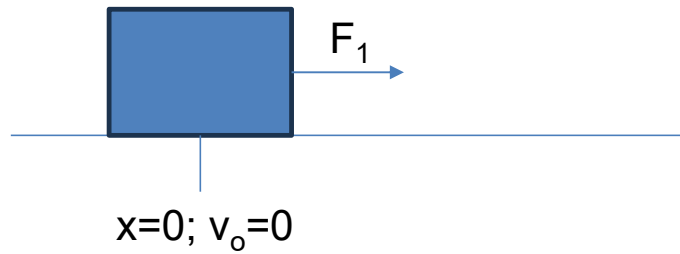


$$\frac{W_1}{W_2} = \frac{F_1}{F_2}$$

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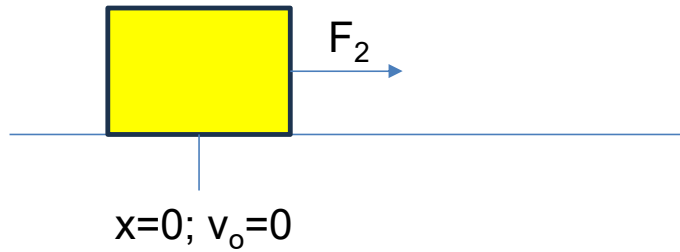
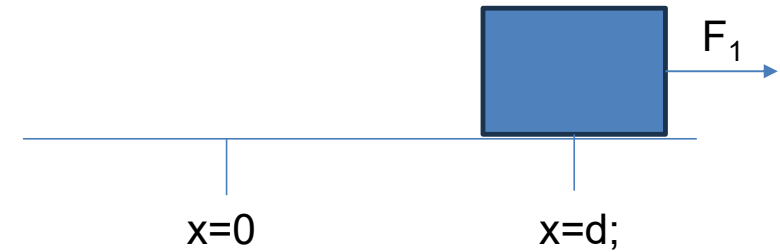
$$\frac{2 \cdot d \cdot m}{t^2} = F$$

$$\frac{W_1}{W_2} = \frac{F_1}{F_2} = \frac{\frac{2 \cdot d \cdot m}{t^2}}{\frac{2 \cdot d \cdot m}{(3t)^2}} = \frac{\frac{1}{t^2}}{\frac{1}{9 \cdot t^2}}$$



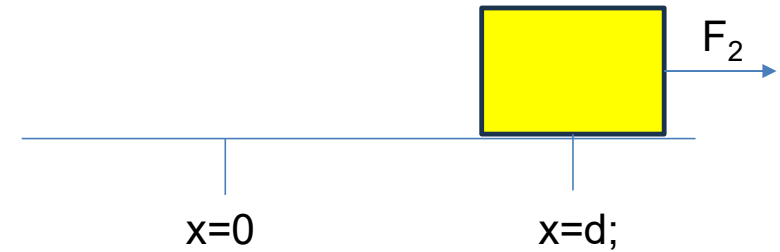
After time= t

Work done = W_1



After time= $3t$

Work done = ?

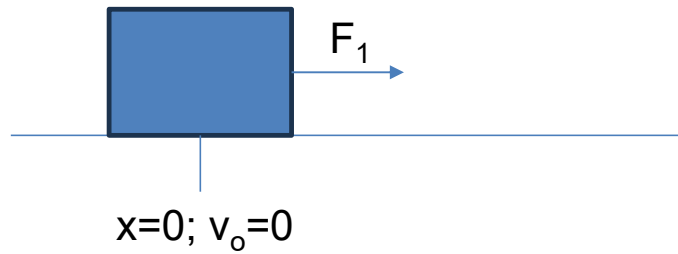


$$\frac{W_1}{W_2} = \frac{F_1}{F_2}$$

$$d = \frac{1}{2} \cdot \frac{F}{m} \cdot t^2$$

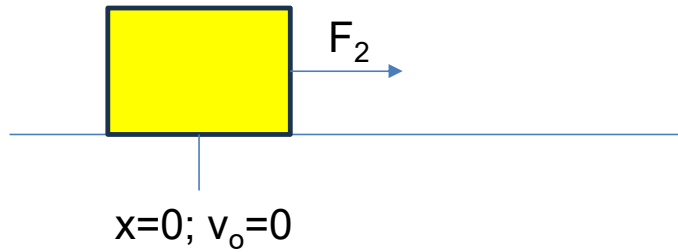
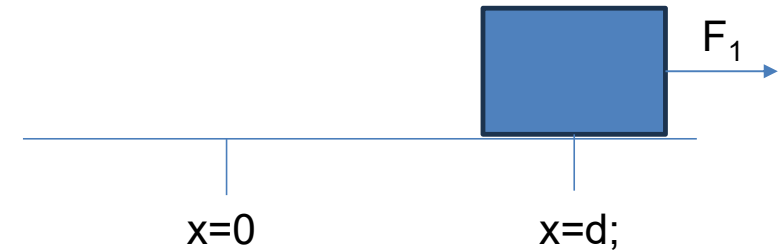
$$\frac{2 \cdot d \cdot m}{t^2} = F$$

$$\frac{W_1}{W_2} = \frac{F_1}{F_2} = \frac{\frac{2 \cdot d \cdot m}{t^2}}{\frac{2 \cdot d \cdot m}{(3t)^2}} = \frac{\frac{1}{t^2}}{\frac{1}{9 \cdot t^2}} = \frac{9t^2}{t^2}$$



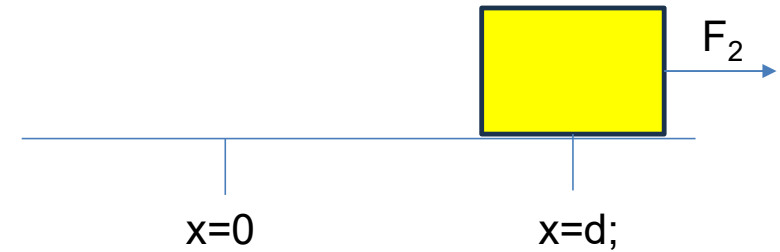
After time= t

Work done = W_1



After time= $3t$

Work done = ?



$$\frac{W_1}{W_2} = \frac{F_1}{F_2}$$

$$d = \frac{1}{2} \cdot \frac{F}{m} \cdot t^2$$

$$\frac{2 \cdot d \cdot m}{t^2} = F$$

$$\frac{W_1}{W_2} = \frac{F_1}{F_2} = \frac{\frac{2 \cdot d \cdot m}{t^2}}{\frac{2 \cdot d \cdot m}{(3t)^2}} = \frac{\frac{1}{t^2}}{\frac{1}{9 \cdot t^2}} = \frac{9t^2}{t^2} = 9$$



10060 Physics (Polytechnical Foundation)

Energy

Kinetic Energy



The kinetic energy is the energy associated with motion

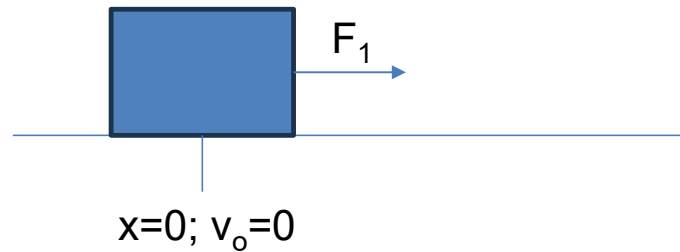
Work-Energy theorem

$$K = \frac{1}{2}mv^2$$

$$\Delta K \equiv K - K_0 = W$$

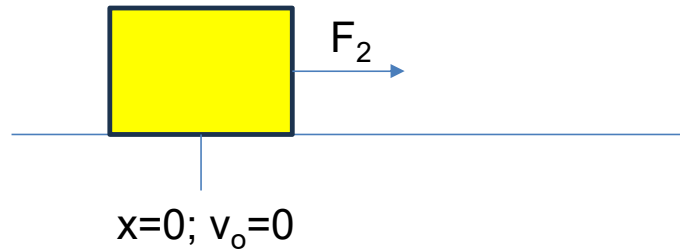
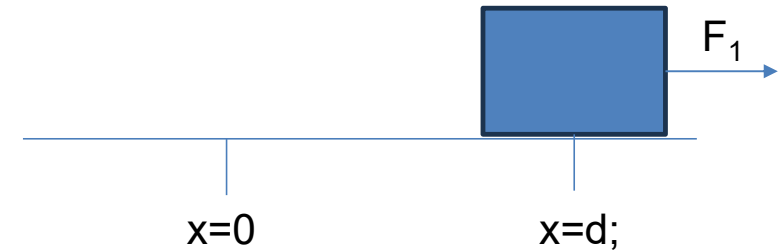
$$W = \vec{F} \cdot \Delta \vec{r}$$

Work is the energy transferred to or from an object due to the action of a force. Positive work is a transfer of energy to the object, and negative work is a transfer of energy from the object.



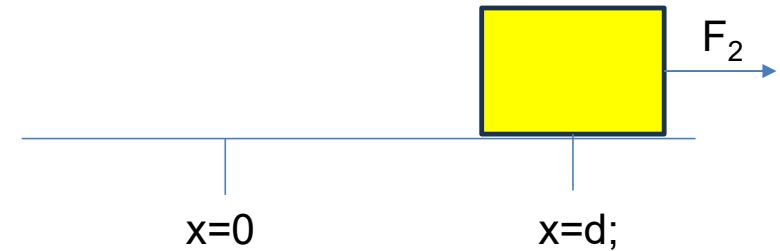
After time= t

Work done = W_1



After time= $3t$

Work done = ?

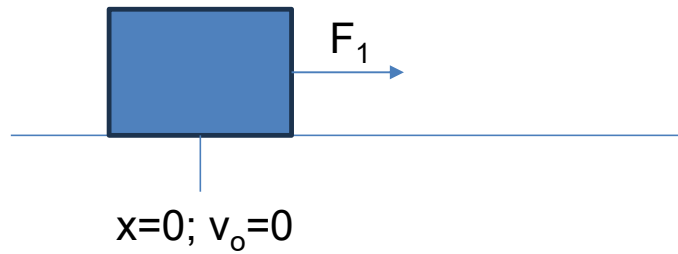


$$\frac{W_1}{W_2} = \frac{F_1}{F_2} = 9$$

$$K = \frac{1}{2}mv^2$$

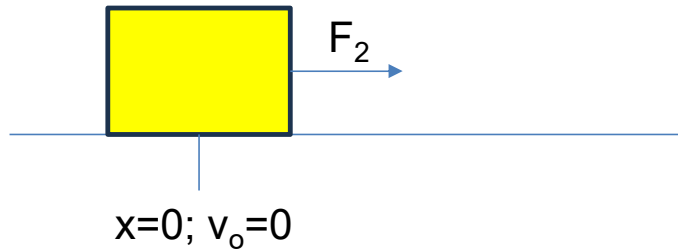
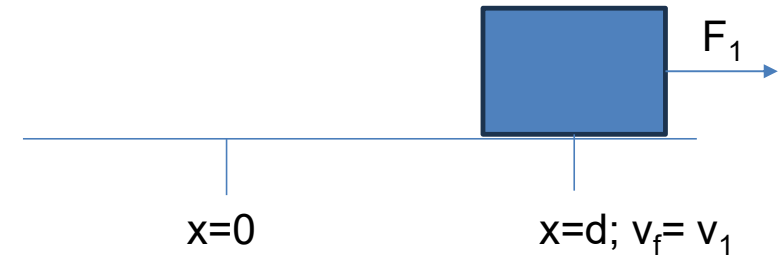
$$\Delta K \equiv K - K_0 = W$$

According to the Work-Energy theorem, the kinetic energy, K , of the box in the first situation should be 9 times larger than in the second situation.



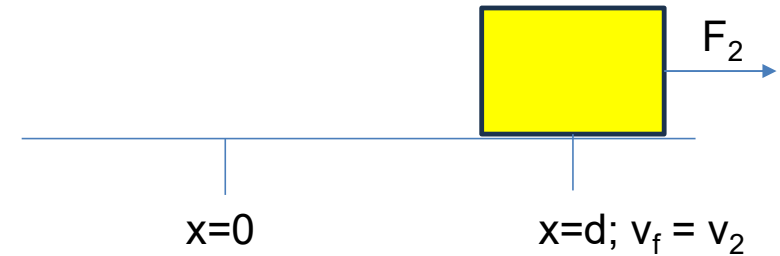
After time= t

Work done = W_1



After time= $3t$

Work done = ?



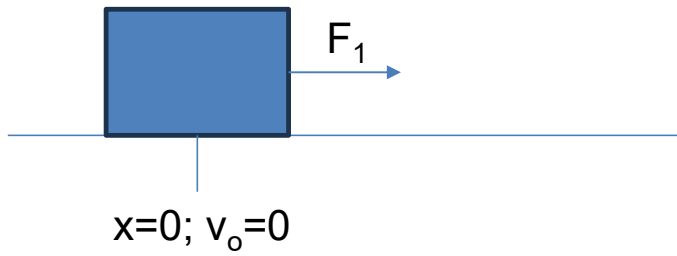
$$\frac{W_1}{W_2} = \frac{F_1}{F_2} = 9$$

$$K = \frac{1}{2}mv^2$$

$$\Delta K \equiv K - K_0 = W$$

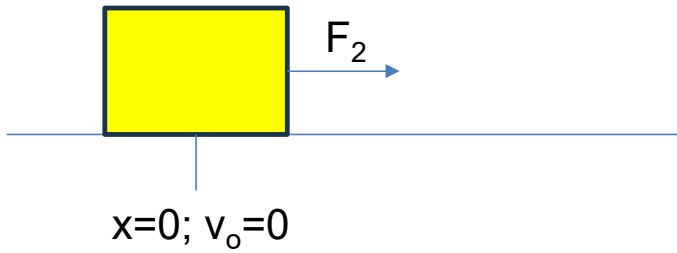
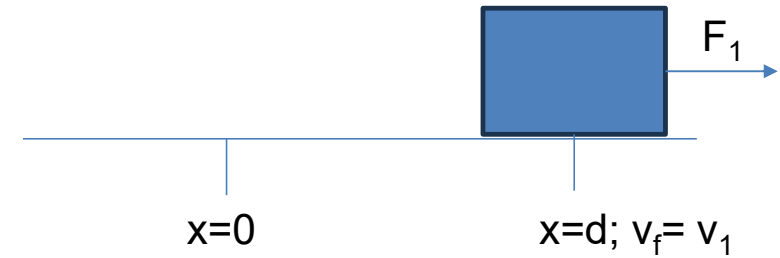
According to the Work-Energy theorem, the kinetic energy, K , of the box in the first situation should be 9 times larger than in the second situation.

Let's check it



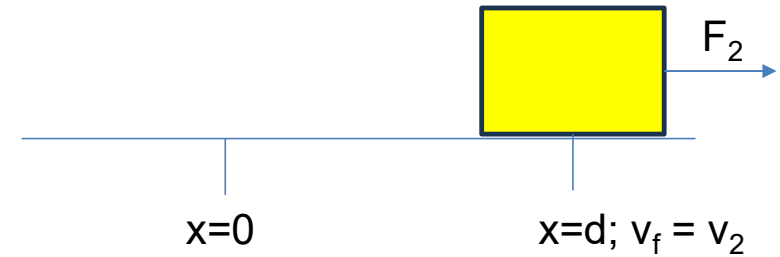
After time= t

Work done = W_1



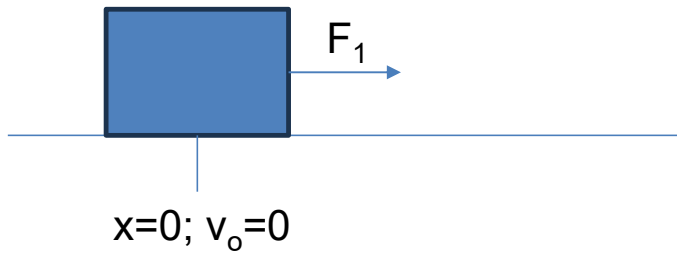
After time= $3t$

Work done = ?



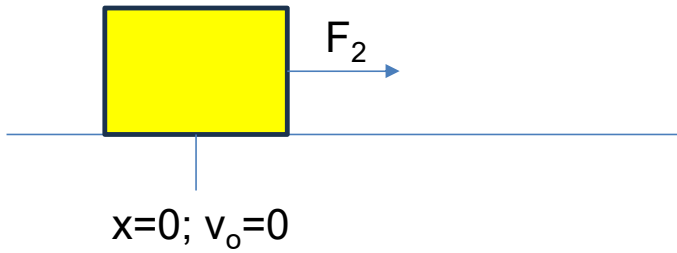
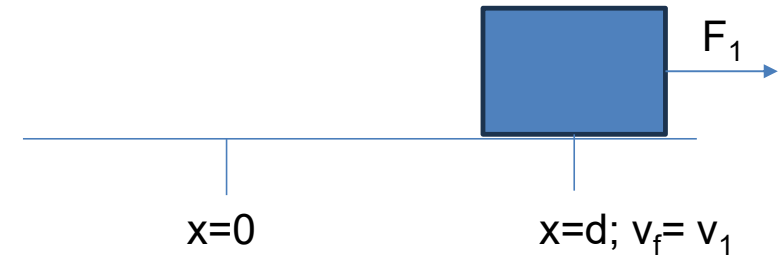
$$K = \frac{1}{2}mv^2$$

$$\frac{W_1}{W_2} = \frac{F_1}{F_2} = 9$$



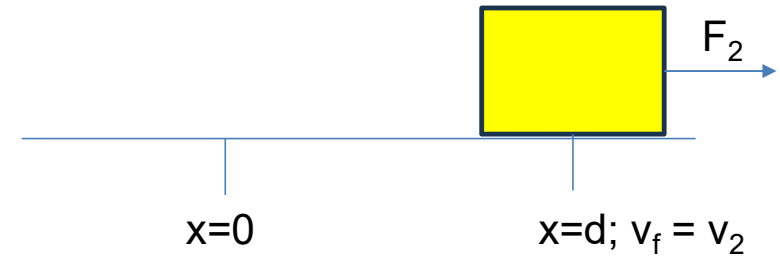
After time= t

Work done = W_1



After time= $3t$

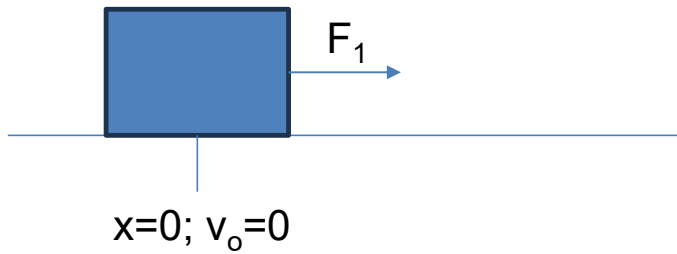
Work done = ?



$$K = \frac{1}{2}mv^2$$

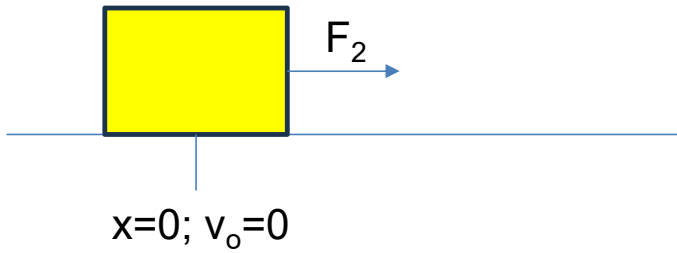
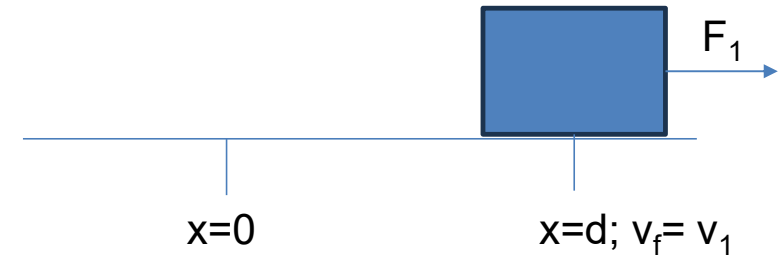
$$\frac{W_1}{W_2} = \frac{F_1}{F_2} = 9$$

$$\frac{K_1}{K_2} = \frac{\frac{1}{2}mv_1^2}{\frac{1}{2}mv_2^2}$$



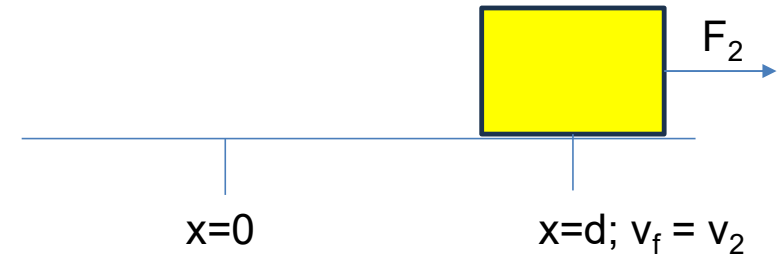
After time= t

Work done = W_1



After time= $3t$

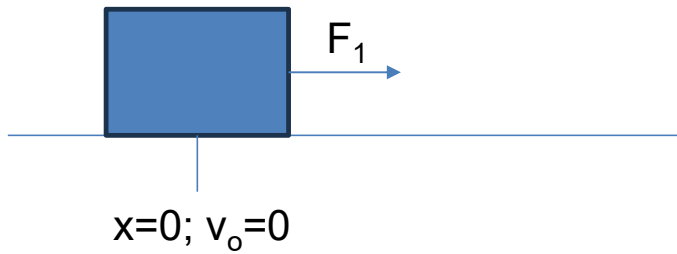
Work done = ?



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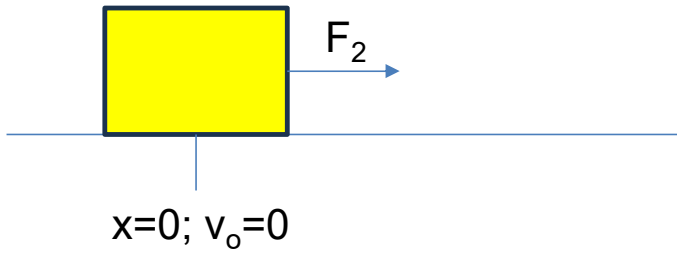
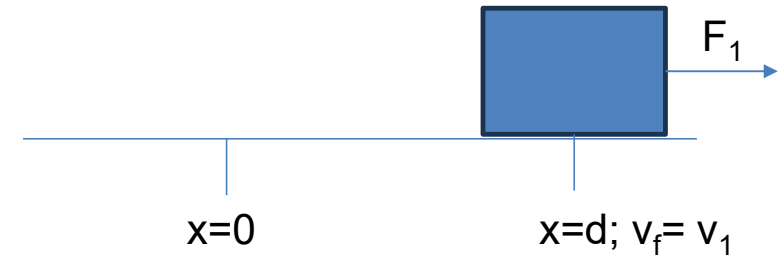
$$\frac{W_1}{W_2} = \frac{F_1}{F_2} = 9$$

$$\frac{K_1}{K_2} = \frac{\frac{1}{2}mv_1^2}{\frac{1}{2}mv_2^2} = \frac{v_1^2}{v_2^2} = 9?$$



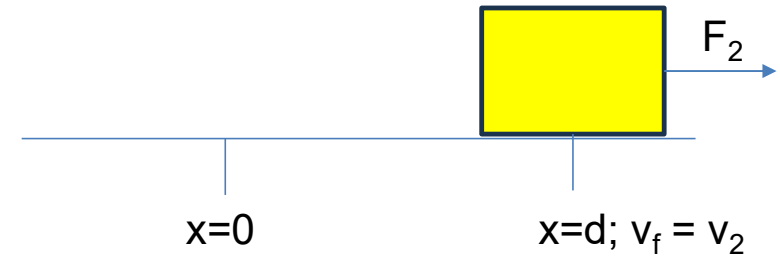
After time= t

Work done = W_1



After time= $3t$

Work done = ?

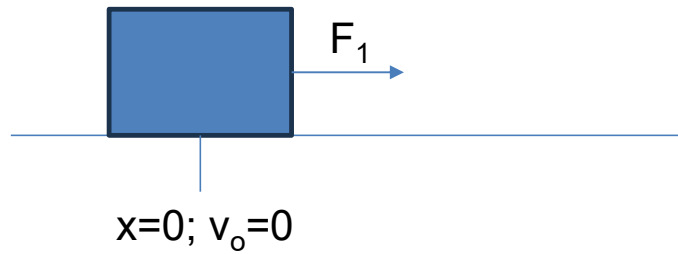


$$K = \frac{1}{2}mv^2$$

$$v = v_o + a \cdot t$$

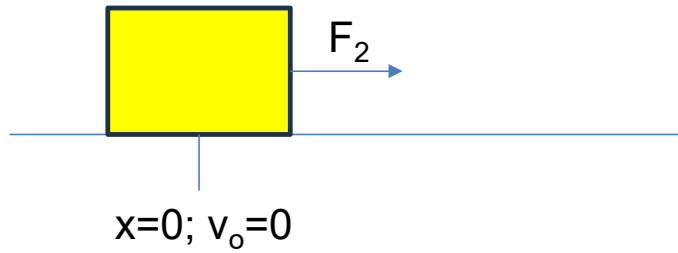
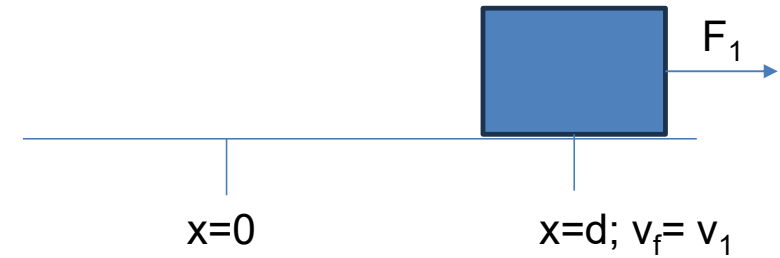
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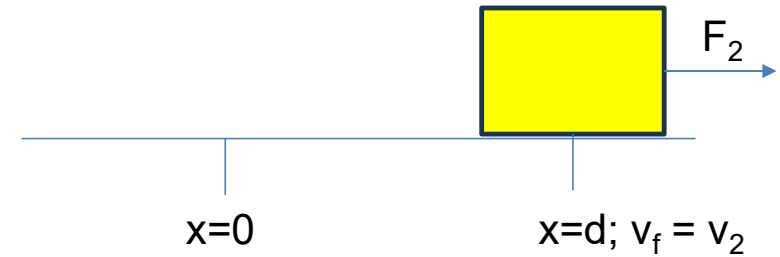
After time= t

Work done = W_1



After time= $3t$

Work done = ?



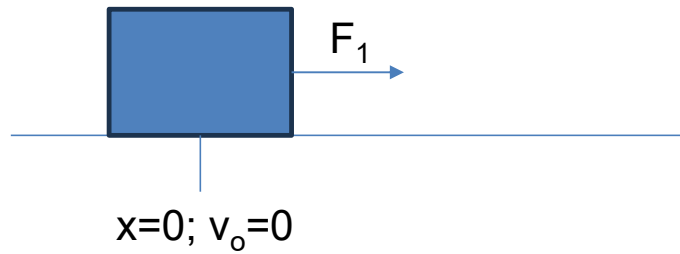
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$$v = v_o + a \cdot t$$

$$v = a \cdot t$$

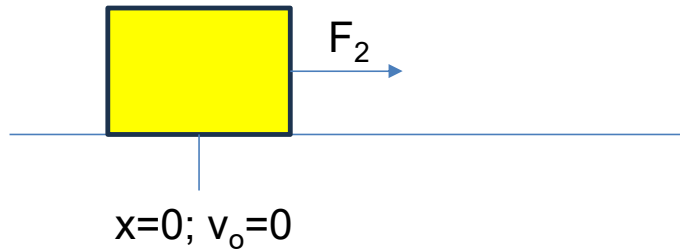
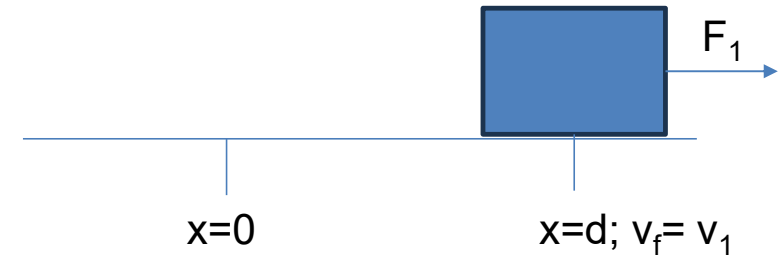
$$\frac{W_1}{W_2} = \frac{F_1}{F_2} = 9$$

$$\frac{K_1}{K_2} = \frac{\frac{1}{2}mv_1^2}{\frac{1}{2}mv_2^2} = \frac{v_1^2}{v_2^2} = 9?$$



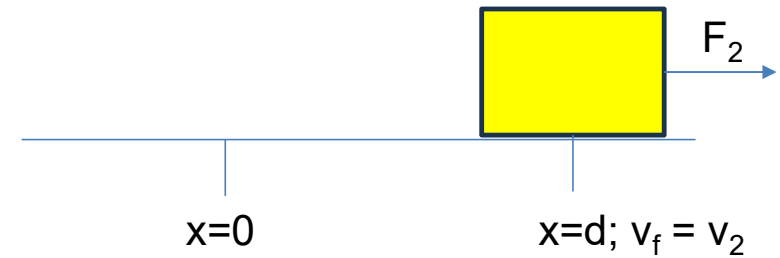
After time= t

Work done = W_1



After time= $3t$

Work done = ?



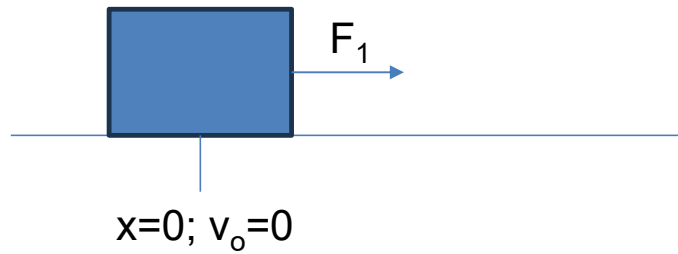
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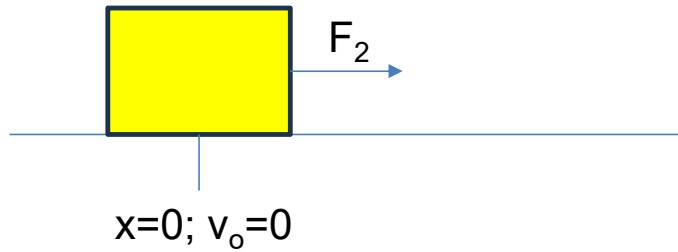
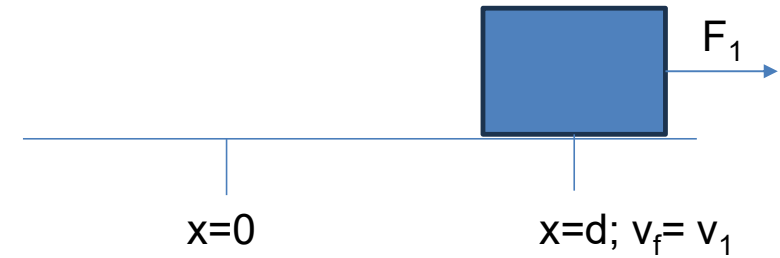
$$v = v_o + a \cdot t$$

$$v = a \cdot t \longrightarrow v = \frac{F}{m} \cdot t$$



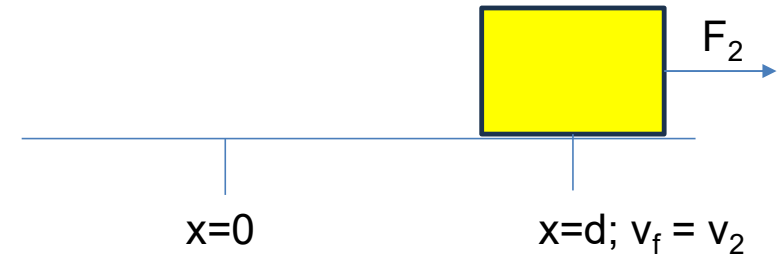
After time= t

Work done = W_1



After time= $3t$

Work done = ?



$$K = \frac{1}{2}mv^2$$

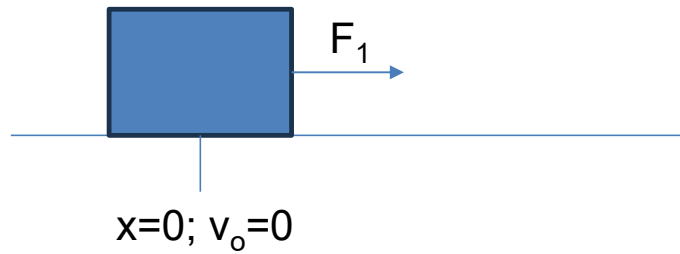
$$\frac{W_1}{W_2} = \frac{F_1}{F_2} = 9$$

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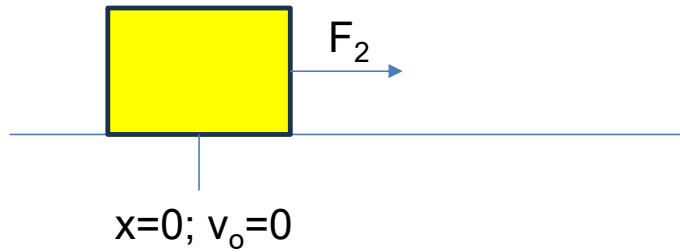
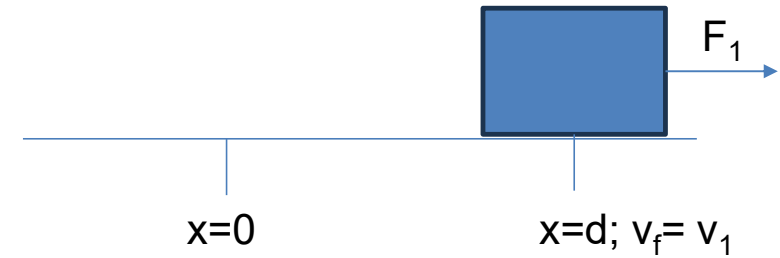
$$v = a \cdot t \longrightarrow v = \frac{F}{m} \cdot t$$

$$v_1 = \frac{F_1}{m} \cdot t$$



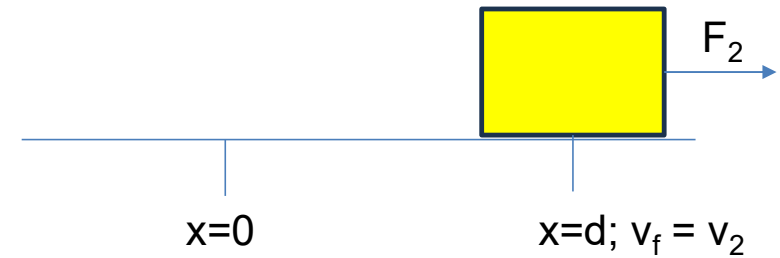
After time= t

Work done = W_1



After time= $3t$

Work done = ?



$$K = \frac{1}{2}mv^2$$

$$\frac{W_1}{W_2} = \frac{F_1}{F_2} = 9$$

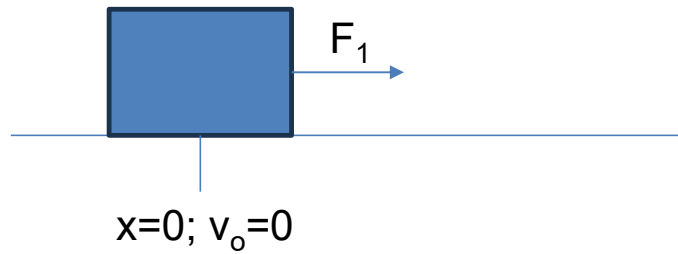
$$\frac{K_1}{K_2} = \frac{\frac{1}{2}mv_1^2}{\frac{1}{2}mv_2^2} = \frac{v_1^2}{v_2^2} = 9?$$

$$v = v_0 + a \cdot t$$

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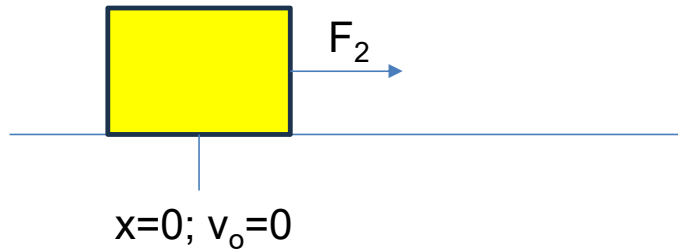
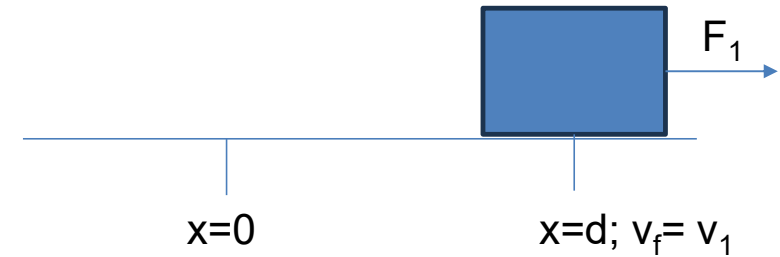
$$v_1 = \frac{F_1}{m} \cdot t$$

$$v_2 = \frac{F_2}{m} \cdot 3t$$



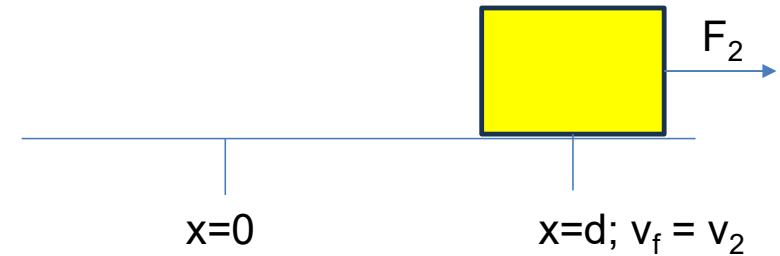
After time= t

Work done = W_1



After time= $3t$

Work done = ?



$$K = \frac{1}{2}mv^2$$

$$\frac{W_1}{W_2} = \frac{F_1}{F_2} = 9$$

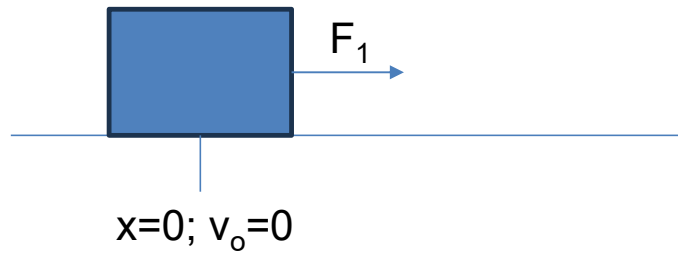
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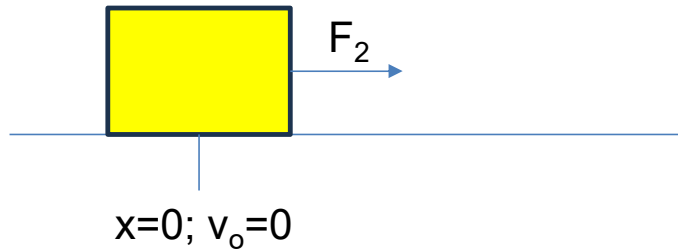
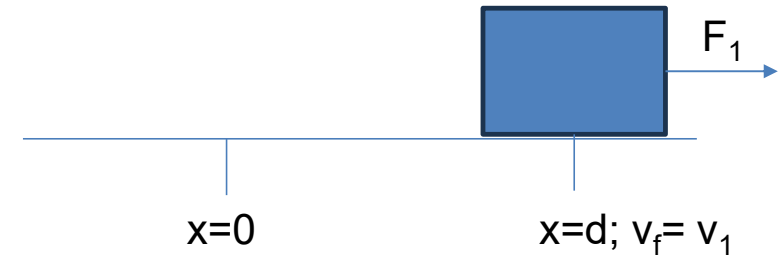
$$v_1 = \frac{F_1}{m} \cdot t$$

$$v_2 = \frac{F_2}{m} \cdot 3t = \frac{F_1}{9} \cdot 3t$$



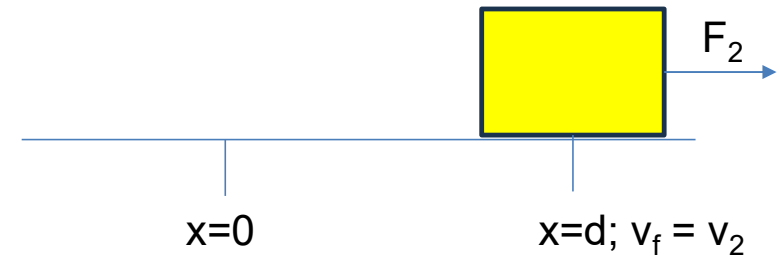
After time= t

Work done = W_1



After time= $3t$

Work done = ?



$$K = \frac{1}{2}mv^2$$

$$\frac{W_1}{W_2} = \frac{F_1}{F_2} = 9$$

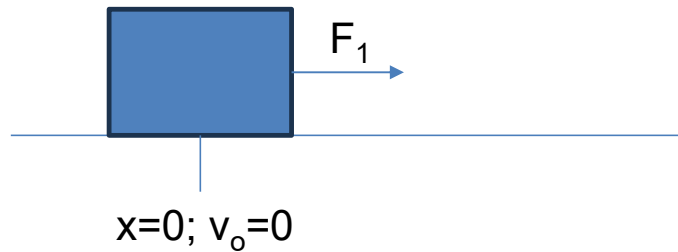
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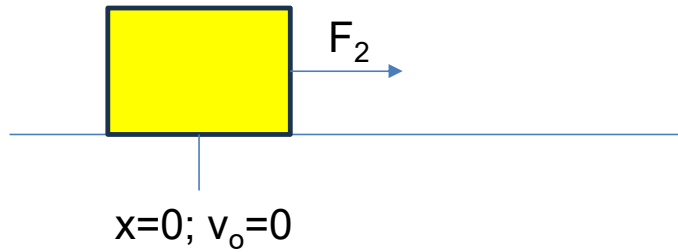
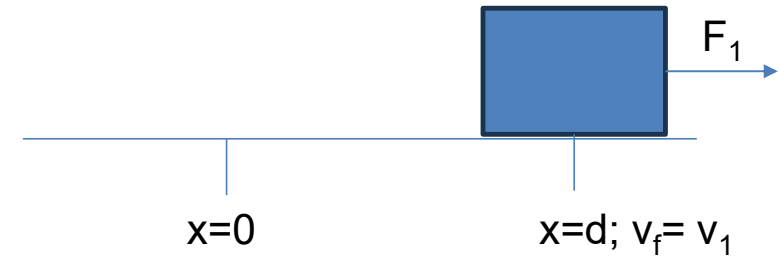
$$v_1 = \frac{F_1}{m} \cdot t$$

$$v_2 = \frac{F_2}{m} \cdot 3t = \frac{\frac{F_1}{9}}{m} \cdot 3t = \frac{F_1}{3m} \cdot t$$



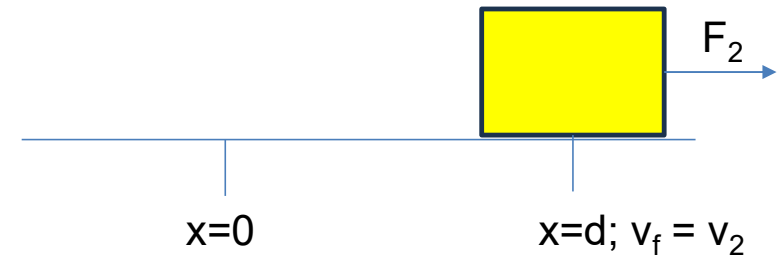
After time= t

Work done = W_1



After time= $3t$

Work done = ?



$$K = \frac{1}{2}mv^2$$

$$\frac{W_1}{W_2} = \frac{F_1}{F_2} = 9$$

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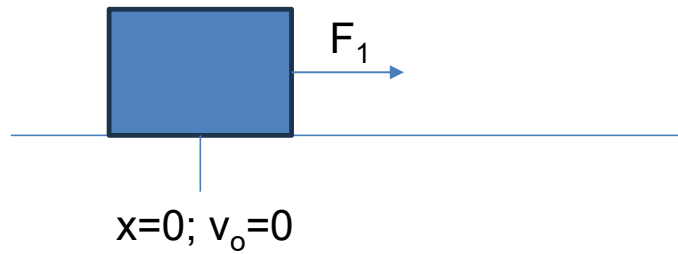
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$$v_1 = \frac{F_1}{m} \cdot t$$

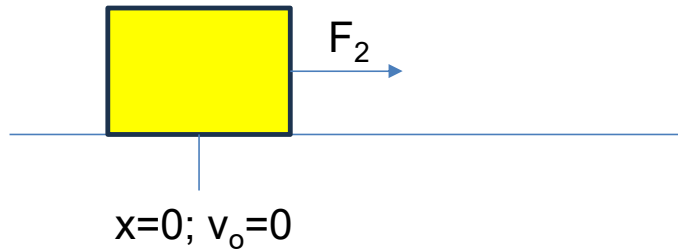
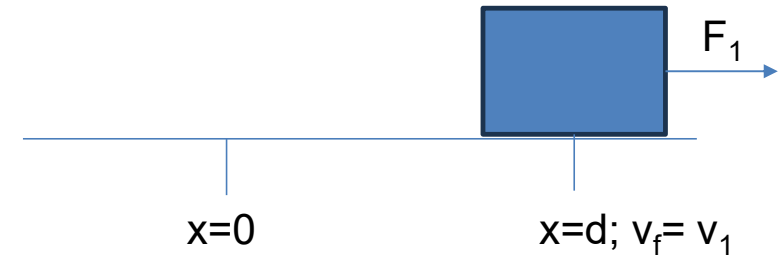
$$v_2 = \frac{F_2}{m} \cdot 3t = \frac{\frac{F_1}{9}}{m} \cdot 3t = \frac{F_1}{3m} \cdot t$$

$$\frac{v_1}{v_2} = 3$$



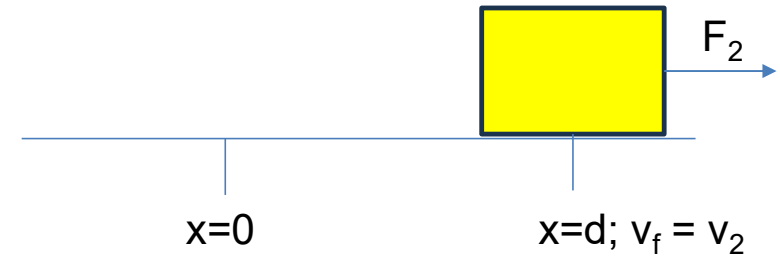
After time= t

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After time= $3t$

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$$v_2 = \frac{F_2}{m} \cdot 3t = \frac{F_1}{9} \cdot 3t = \frac{F_1}{3} \cdot t$$

$$\frac{v_1}{v_2} = 3$$

$$\frac{v_1^2}{v_2^2} = 9$$

10060 Physics (Polytechnical Foundation)

Power

Power definition

Average power during time interval Δt $\rightarrow P_{\text{av}} = \frac{\Delta W}{\Delta t}$ $\left\{ \begin{array}{l} \Delta W \text{ Work done during time interval} \\ \Delta t \text{ Duration of time interval} \end{array} \right.$

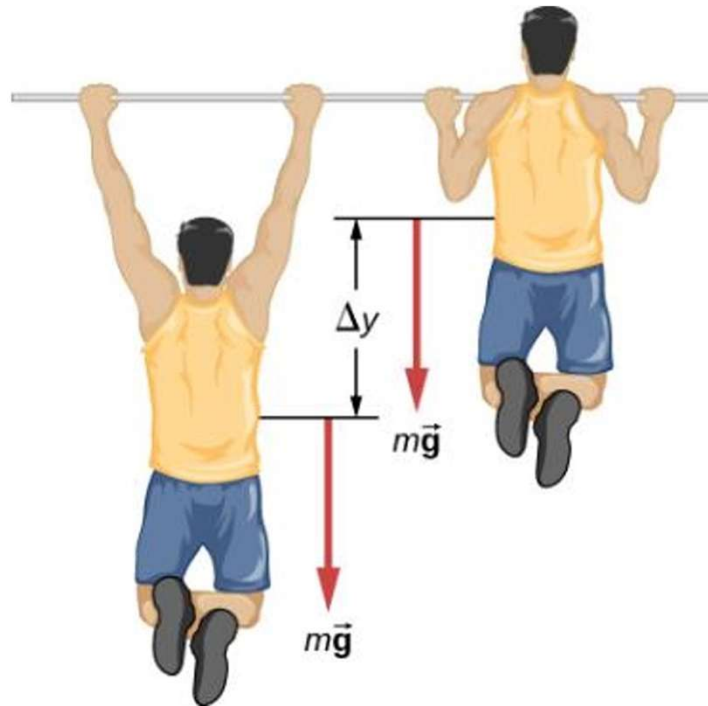
Instantaneous power $\rightarrow P = \lim_{\Delta t \rightarrow 0} \frac{\Delta W}{\Delta t} = \frac{dW}{dt}$ $\left\{ \begin{array}{l} \text{Time rate of} \\ \text{doing work} \end{array} \right.$

Average power over infinitesimally short time interval

$$W = \vec{F} \cdot \Delta \vec{r}$$

Instantaneous power for a force doing work on a particle $\rightarrow P = \vec{F} \cdot \vec{v}$ $\left\{ \begin{array}{l} \vec{F} \text{ Force that acts on particle} \\ \vec{v} \text{ Velocity of particle} \end{array} \right.$

Power: pull-ups



What is the power of doing 1 pull-ups in 1 sec?
(Discuss in groups)

Summary: Kinetic Energy, Work, Power

$$K = \frac{1}{2}mv^2$$

$$\Delta K \equiv K - K_0 = W$$

$$W = \vec{F} \cdot \Delta \vec{r}$$

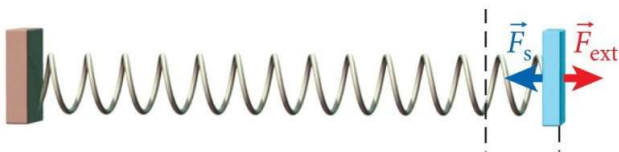
Work is the energy transferred to or from an object due to the action of a force. Positive work is a transfer of energy to the object, and negative work is a transfer of energy from the object.

$$F_s = -kx$$

$$W = -W_s = \frac{1}{2}kx_c^2$$

$$P = \frac{dW}{dt}$$

$$\bar{P} = \frac{W}{\Delta t}$$



$$P = \vec{F} \cdot \vec{v}$$

